Experiment 9 - Generative Adversarial Networks

Dataset: Fashion-MNIST | **Models:** Vanilla GAN, DCGAN

```
!pip install wandb huggingface_hub -q
```

```
import wandb
wandb.login()
```

```
/usr/local/lib/python3.12/dist-packages/notebook/notebookapp.py:191: SyntaxWarning: invalid escape sequence '\/'
  | |_| | '_ \/ _` / _` |  _/ -_)
wandb: (1) Create a W&B account
wandb: (2) Use an existing W&B account
wandb: (3) Don't visualize my results
wandb: Enter your choice: 2
wandb: You chose 'Use an existing W&B account'
wandb: Logging into https://api.wandb.ai. (Learn how to deploy a W&B server locally: https://wandb.me/wandb-server)
wandb: Create a new API key at: https://wandb.ai/authorize?ref=models
wandb: Store your API key securely and do not share it.
wandb: Paste your API key and hit enter: ··********
wandb: No netrc file found, creating one.
wandb: Appending key for api.wandb.ai to your netrc file: /root/.netrc
wandb: Currently logged in as: mayankjain_25rco06 (mayankjain_25rco06-delhi-technological-university) to https://api.wandb.ai. Use `wandb login --
True
```

```
from huggingface_hub import notebook_login
notebook_login()
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
```

Illegal header value b'Bearer '

```
import torch
import torch.nn as nn
```

```
import torch.optim as optim
import torchvision
import torchvision.transforms as transforms
import matplotlib.pyplot as plt
import numpy as np
import os
from torch.utils.data import DataLoader
from huggingface_hub import HfApi, create_repo

# change to your huggingface username
HF_REPO = "Mayank2424/exp9-gans"

WANDB_PROJECT = "exp9-gans"
EPOCHS        = 10
BATCH_SIZE    = 128
LATENT_DIM    = 100
DEVICE        = torch.device("cuda" if torch.cuda.is_available() else "cpu")

LOSS_FNS   = ["bce", "lsgan", "wgan"]
OPTIMIZERS = ["adam", "sgd", "rmsprop"]
MODELS     = ["vanilla", "dcgan"]

print(DEVICE)
```

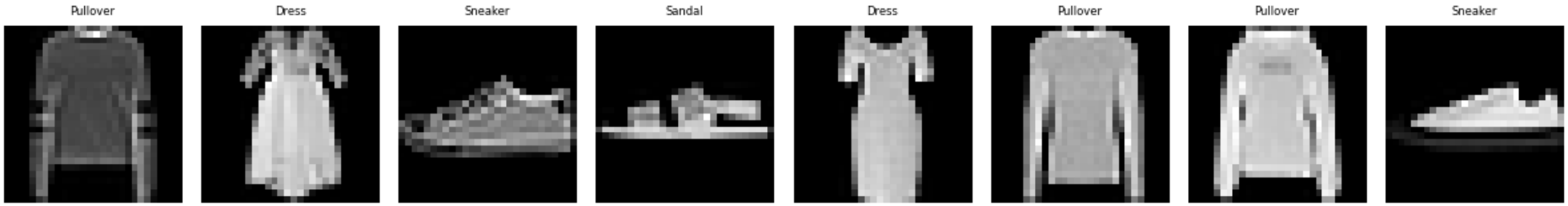
```
cuda
```

Dataset

```
# normalize to [-1, 1] for GAN training
transform = transforms.Compose([
    transforms.ToTensor(),
    transforms.Normalize([0.5], [0.5])
])

train_data = torchvision.datasets.FashionMNIST(root='./data', train=True,  download=True, transform=transform)
test_data  = torchvision.datasets.FashionMNIST(root='./data', train=False, download=True, transform=transform)

train_loader = DataLoader(train_data, batch_size=BATCH_SIZE, shuffle=True,  num_workers=2)
test_loader  = DataLoader(test_data,  batch_size=BATCH_SIZE, shuffle=False, num_workers=2)

print(f'train: {len(train_data)}  test: {len(test_data)}')

imgs, labels = next(iter(train_loader))
```

```
class_names = ['T-shirt','Trouser','Pullover','Dress','Coat','Sandal','Shirt','Sneaker','Bag','Ankle boot']
fig, axes = plt.subplots(1, 8, figsize=(12, 2))
for i in range(8):
    axes[i].imshow(imgs[i].squeeze(), cmap='gray')
    axes[i].set_title(class_names[labels[i]], fontsize=6)
    axes[i].axis('off')
plt.tight_layout()
plt.show()
```

```
100%|██████████| 26.4M/26.4M [00:02<00:00, 12.9MB/s]
100%|██████████| 29.5k/29.5k [00:00<00:00, 200kB/s]
100%|██████████| 4.42M/4.42M [00:01<00:00, 3.77MB/s]
100%|██████████| 5.15k/5.15k [00:00<00:00, 12.0MB/s]train: 60000  test: 10000
```



Vanilla GAN - Generator and Discriminator

```
class VanillaGenerator(nn.Module):
    def __init__(self, latent_dim):
        super(VanillaGenerator, self).__init__()
        self.net = nn.Sequential(
            nn.Linear(latent_dim, 256),
            nn.LeakyReLU(0.2),
            nn.Linear(256, 512),
            nn.LeakyReLU(0.2),
            nn.Linear(512, 1024),
            nn.LeakyReLU(0.2),
            nn.Linear(1024, 784),
            nn.Tanh()
        )

    def forward(self, z):
        return self.net(z).view(-1, 1, 28, 28)
```

```
class VanillaDiscriminator(nn.Module):
    def __init__(self):
        super(VanillaDiscriminator, self).__init__()
        self.net = nn.Sequential(
            nn.Flatten(),
            nn.Linear(784, 512),
            nn.LeakyReLU(0.2),
            nn.Linear(512, 256),
            nn.LeakyReLU(0.2),
            nn.Linear(256, 1)
        )

    def forward(self, x):
        return self.net(x)
```

DCGAN - Generator and Discriminator

```
class DCGenerator(nn.Module):
    def __init__(self, latent_dim):
        super(DCGenerator, self).__init__()
        self.net = nn.Sequential(
            # latent -> 7x7x256
            nn.ConvTranspose2d(latent_dim, 256, 7, 1, 0, bias=False),
            nn.BatchNorm2d(256),
            nn.ReLU(True),
            # 7x7 -> 14x14
            nn.ConvTranspose2d(256, 128, 4, 2, 1, bias=False),
            nn.BatchNorm2d(128),
            nn.ReLU(True),
            # 14x14 -> 28x28
            nn.ConvTranspose2d(128, 1, 4, 2, 1, bias=False),
            nn.Tanh()
        )

    def forward(self, z):
        z = z.view(z.size(0), z.size(1), 1, 1)
        return self.net(z)


class DCDiscriminator(nn.Module):
    def __init__(self):
        super(DCDiscriminator, self).__init__()
```

```
        self.net = nn.Sequential(
            # 28x28 -> 14x14
            nn.Conv2d(1, 64, 4, 2, 1, bias=False),
            nn.LeakyReLU(0.2, inplace=True),
            # 14x14 -> 7x7
            nn.Conv2d(64, 128, 4, 2, 1, bias=False),
            nn.BatchNorm2d(128),
            nn.LeakyReLU(0.2, inplace=True),
            # 7x7 -> 1x1
            nn.Conv2d(128, 1, 7, 1, 0, bias=False)
        )

    def forward(self, x):
        return self.net(x).view(-1, 1)
```

Loss Functions - BCE, LSGAN, WGAN

```
bce = nn.BCEWithLogitsLoss()
mse = nn.MSELoss()

def disc_loss(real_out, fake_out, loss_type):
    if loss_type == 'bce':
        real_loss = bce(real_out, torch.ones_like(real_out))
        fake_loss = bce(fake_out, torch.zeros_like(fake_out))
        return (real_loss + fake_loss) / 2
    elif loss_type == 'lsgan':
        real_loss = mse(real_out, torch.ones_like(real_out))
        fake_loss = mse(fake_out, torch.zeros_like(fake_out))
        return (real_loss + fake_loss) / 2
    elif loss_type == 'wgan':
        return -(real_out.mean() - fake_out.mean())

def gen_loss(fake_out, loss_type):
    if loss_type == 'bce':
        return bce(fake_out, torch.ones_like(fake_out))
    elif loss_type == 'lsgan':
        return mse(fake_out, torch.ones_like(fake_out))
    elif loss_type == 'wgan':
        return -fake_out.mean()
```

Optimizer Factory

```
def get_optimizer(model, name):
    if name == 'adam':
        # betas tuned for GAN stability
        return optim.Adam(model.parameters(), lr=2e-4, betas=(0.5, 0.999))
    elif name == 'sgd':
        return optim.SGD(model.parameters(), lr=1e-3, momentum=0.9)
    else:
        return optim.RMSprop(model.parameters(), lr=5e-5)
```

Training Loop

```
def train_epoch(G, D, loader, opt_g, opt_d, loss_type):
    G.train(); D.train()
    g_running = 0
    d_running = 0

    for real, _ in loader:
        real = real.to(DEVICE)
        bs   = real.size(0)

        # train discriminator
        z         = torch.randn(bs, LATENT_DIM).to(DEVICE)
        fake      = G(z).detach()
        real_out  = D(real)
        fake_out  = D(fake)
        d_loss    = disc_loss(real_out, fake_out, loss_type)
        opt_d.zero_grad()
        d_loss.backward()
        opt_d.step()

        # weight clipping for wgan
        if loss_type == 'wgan':
            for p in D.parameters():
                p.data.clamp_(-0.01, 0.01)

        # train generator
        z        = torch.randn(bs, LATENT_DIM).to(DEVICE)
        fake     = G(z)
        fake_out = D(fake)
        g_loss   = gen_loss(fake_out, loss_type)
        opt_g.zero_grad()
        g_loss.backward()
```

```
            opt_g.step()

            g_running += g_loss.item()
            d_running += d_loss.item()

        n = len(loader)
        return g_running / n, d_running / n
```

Visualization

```
def show_generated(G, epoch, tag, n=16):
    G.eval()
    with torch.no_grad():
        z    = torch.randn(n, LATENT_DIM).to(DEVICE)
        imgs = G(z).cpu()
    imgs = (imgs + 1) / 2  # [-1,1] -> [0,1]
    fig, axes = plt.subplots(2, 8, figsize=(12, 3))
    for i, ax in enumerate(axes.flatten()):
        ax.imshow(imgs[i].squeeze(), cmap='gray')
        ax.axis('off')
    plt.suptitle(f'epoch {epoch} - {tag}', fontsize=9)
    plt.tight_layout()
    plt.show()
    return fig
```

Main Experiment Runner

```
results       = {}
saved_generators = {}

def run_experiment(arch, loss_type, opt_name):
    name = f'{arch}_{loss_type}_{opt_name}'
    print(f'\nstarting: {name}')

    wandb.init(project=WANDB_PROJECT, name=name, config={
        'arch': arch, 'loss': loss_type, 'optimizer': opt_name, 'epochs': EPOCHS
    })

    if arch == 'vanilla':
        G = VanillaGenerator(LATENT_DIM).to(DEVICE)
        D = VanillaDiscriminator().to(DEVICE)
```

```
        else:
            G = DCGenerator(LATENT_DIM).to(DEVICE)
            D = DCDiscriminator().to(DEVICE)

        opt_g = get_optimizer(G, opt_name)
        opt_d = get_optimizer(D, opt_name)

        g_losses = []
        d_losses = []

        for epoch in range(1, EPOCHS + 1):
            g_loss, d_loss = train_epoch(G, D, train_loader, opt_g, opt_d, loss_type)
            g_losses.append(g_loss)
            d_losses.append(d_loss)
            wandb.log({'epoch': epoch, 'g_loss': g_loss, 'd_loss': d_loss})
            print(f'  epoch {epoch}/{EPOCHS}  g_loss={g_loss:.4f}  d_loss={d_loss:.4f}')

            if epoch % 5 == 0 or epoch == EPOCHS:
                fig = show_generated(G, epoch, name)
                wandb.log({f'generated_ep{epoch}': wandb.Image(fig)})
                plt.close(fig)

        # loss curves
        fig, ax = plt.subplots(figsize=(7, 3))
        ax.plot(g_losses, label='generator')
        ax.plot(d_losses, label='discriminator')
        ax.set_xlabel('epoch')
        ax.set_ylabel('loss')
        ax.set_title(name)
        ax.legend()
        plt.tight_layout()
        plt.show()
        wandb.log({'loss_curve': wandb.Image(fig)})
        plt.close(fig)

        torch.save(G.state_dict(), f'{name}_G.pt')
        torch.save(D.state_dict(), f'{name}_D.pt')
        saved_generators[name] = G
        results[name] = {'g_loss': g_losses[-1], 'd_loss': d_losses[-1]}

        wandb.finish()
        return G, D
```

Quick Test - run this first

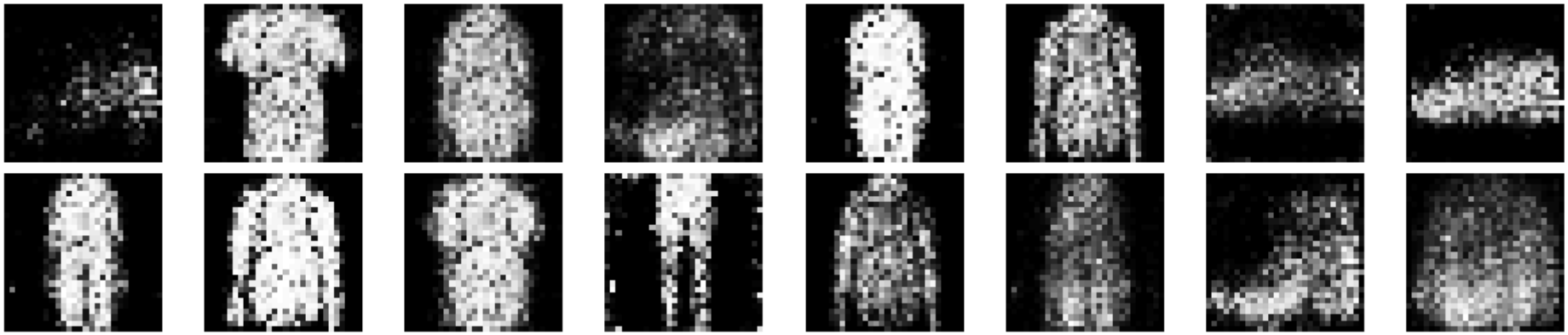
```
run_experiment('vanilla', 'bce', 'adam')
run_experiment('dcgan',   'bce', 'adam')
```

```
starting: vanilla_bce_adam
```

Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_081623-qctobhup`
Syncing run **vanilla_bce_adam** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/qctobhup

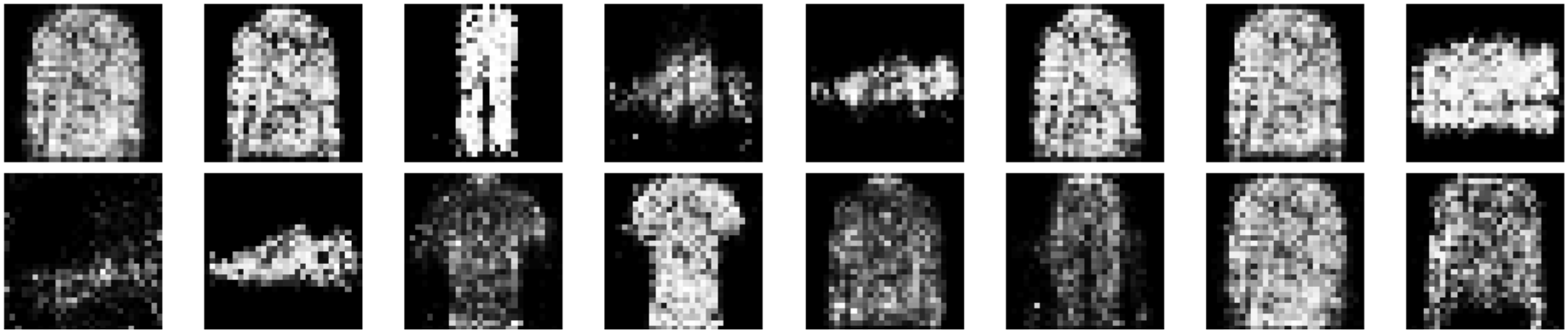
```
epoch 1/10  g_loss=1.1580  d_loss=0.4966
epoch 2/10  g_loss=1.8737  d_loss=0.4712
epoch 3/10  g_loss=1.8037  d_loss=0.4777
epoch 4/10  g_loss=1.5137  d_loss=0.5316
epoch 5/10  g_loss=1.4039  d_loss=0.5439
```

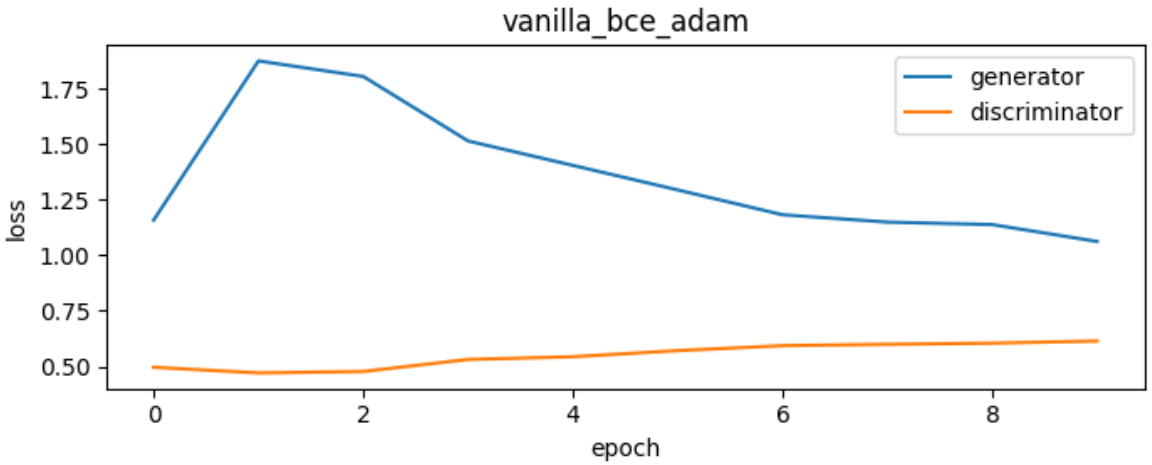
epoch 5 - vanilla_bce_adam



```
epoch 6/10  g_loss=1.2936  d_loss=0.5714
epoch 7/10  g_loss=1.1820  d_loss=0.5938
epoch 8/10  g_loss=1.1491  d_loss=0.5999
epoch 9/10  g_loss=1.1378  d_loss=0.6049
epoch 10/10  g_loss=1.0621  d_loss=0.6146
```

epoch 10 - vanilla_bce_adam





Run history:

d_loss
epoch
g_loss

Run summary:

d_loss 0.61462
epoch 10
g_loss 1.06211

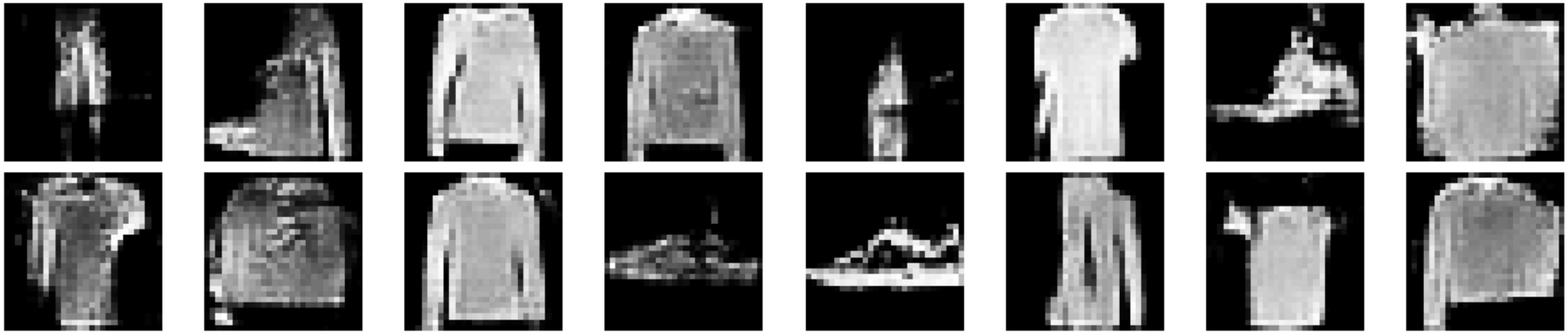
View run **vanilla_bce_adam** at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/qctobhup
View project at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: `./wandb/run-20260416_081623-qctobhup/logs`

```
starting: dcgan_bce_adam
```
Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_081857-tad2cpry`
Syncing run **dcgan_bce_adam** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/tad2cpry

```
epoch 1/10  g_loss=1.7849  d_loss=0.3360
epoch 2/10  g_loss=1.4380  d_loss=0.3787
epoch 3/10  g_loss=1.2793  d_loss=0.4455
```

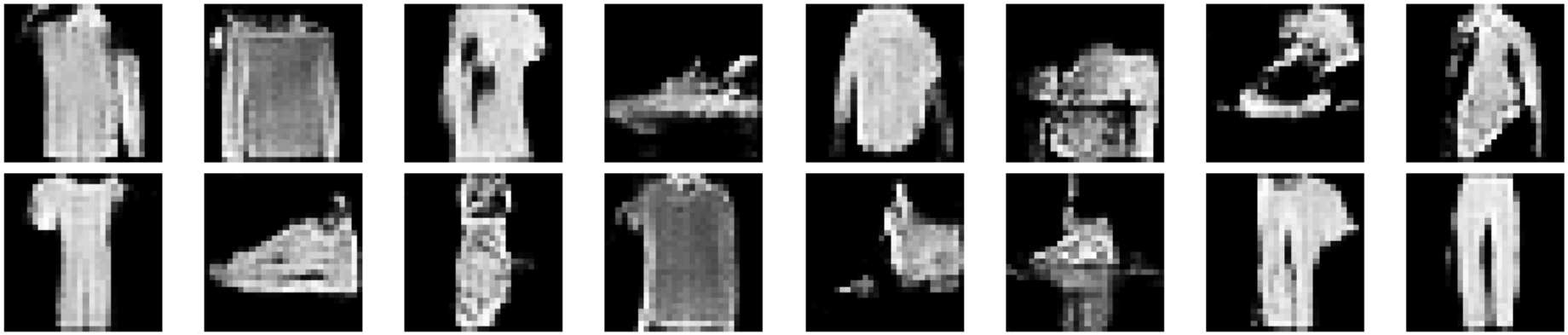
```
epoch 4/10  g_loss=1.2005  d_loss=0.4782
epoch 5/10  g_loss=1.2021  d_loss=0.4798
```

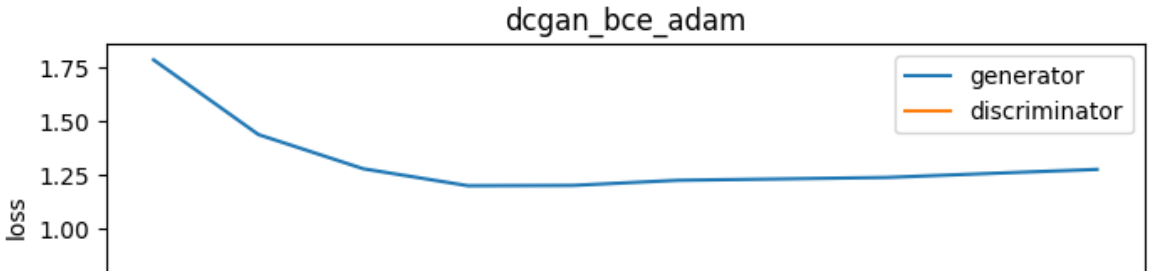
epoch 5 - dcgan_bce_adam

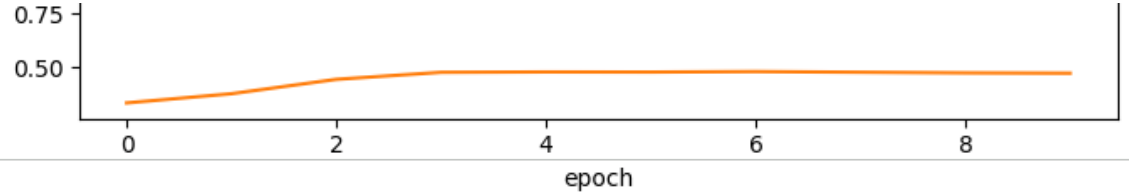


```
epoch 6/10  g_loss=1.2259  d_loss=0.4793
epoch 7/10  g_loss=1.2322  d_loss=0.4815
epoch 8/10  g_loss=1.2391  d_loss=0.4785
epoch 9/10  g_loss=1.2581  d_loss=0.4755
epoch 10/10  g_loss=1.2765  d_loss=0.4741
```

epoch 10 - dcgan_bce_adam







Run history:

d_loss
epoch
g_loss

Run summary:

d_loss 0.47407
epoch 10
g_loss 1.27646

View run **dcgan_bce_adam** at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/tad2cpry
View project at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: `./wandb/run-20260416_081857-tad2cpry/logs`

```
(DCGenerator(
   (net): Sequential(
     (0): ConvTranspose2d(100, 256, kernel_size=(7, 7), stride=(1, 1), bias=False)
     (1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
     (2): ReLU(inplace=True)
     (3): ConvTranspose2d(256, 128, kernel_size=(4, 4), stride=(2, 2), padding=(1, 1), bias=False)
     (4): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
     (5): ReLU(inplace=True)
     (6): ConvTranspose2d(128, 1, kernel_size=(4, 4), stride=(2, 2), padding=(1, 1), bias=False)
     (7): Tanh()
   )
 ),
 DCDiscriminator(
   (net): Sequential(
     (0): Conv2d(1, 64, kernel_size=(4, 4), stride=(2, 2), padding=(1, 1), bias=False)
     (1): LeakyReLU(negative_slope=0.2, inplace=True)
     (2): Conv2d(64, 128, kernel_size=(4, 4), stride=(2, 2), padding=(1, 1), bias=False)
     (3): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
     (4): LeakyReLU(negative_slope=0.2, inplace=True)
     (5): Conv2d(128, 1, kernel_size=(7, 7), stride=(1, 1), bias=False)
```

Full Grid - all combinations

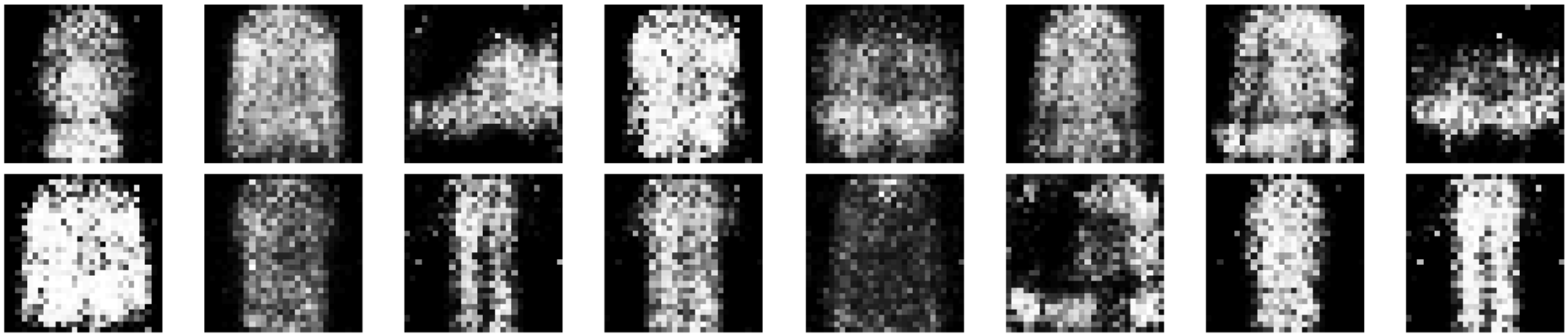
```
for arch in MODELS:
    for lf in LOSS_FNS:
        for opt in OPTIMIZERS:
            run_experiment(arch, lf, opt)
```

```
starting: vanilla_bce_adam
```

Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_082151-rx4nvt47`
Syncing run **vanilla_bce_adam** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/rx4nvt47

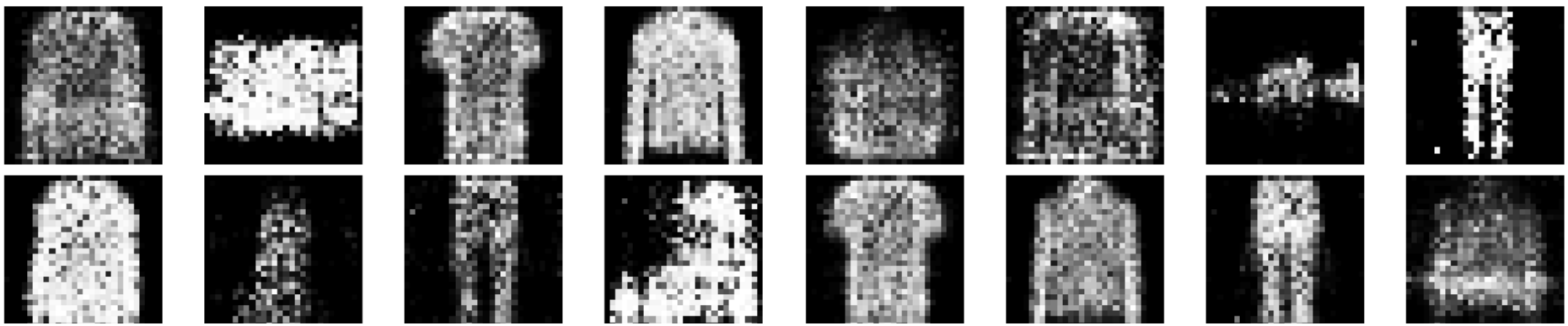
```
epoch 1/10  g_loss=1.1623  d_loss=0.4855
epoch 2/10  g_loss=2.0177  d_loss=0.4075
epoch 3/10  g_loss=1.9575  d_loss=0.4382
epoch 4/10  g_loss=1.8077  d_loss=0.4964
epoch 5/10  g_loss=1.3339  d_loss=0.5569
```

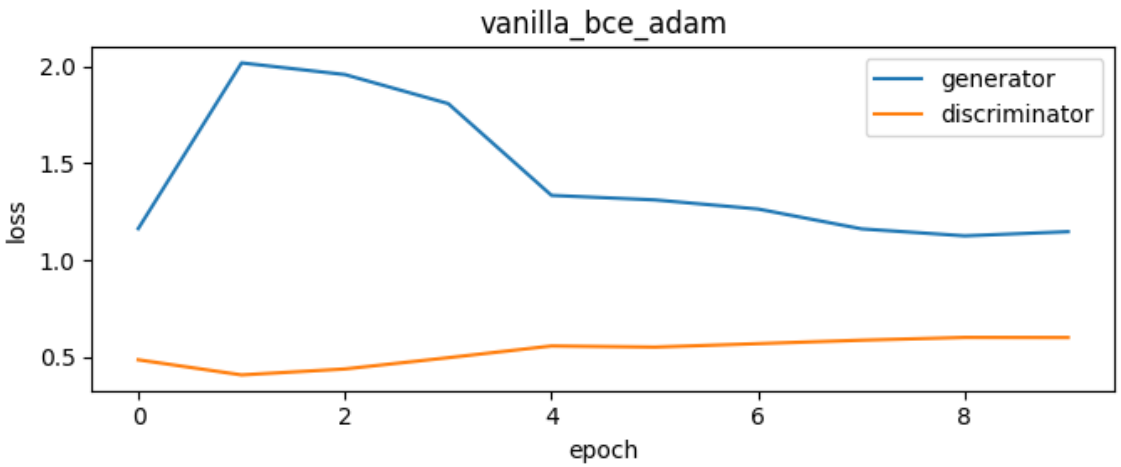
epoch 5 - vanilla_bce_adam



```
epoch 6/10  g_loss=1.3107  d_loss=0.5510
epoch 7/10  g_loss=1.2635  d_loss=0.5689
epoch 8/10  g_loss=1.1612  d_loss=0.5867
epoch 9/10  g_loss=1.1251  d_loss=0.6009
epoch 10/10  g_loss=1.1466  d_loss=0.6003
```

epoch 10 - vanilla_bce_adam





Run history:

d_loss
epoch
g_loss

Run summary:

d_loss 0.60031
epoch 10
g_loss 1.14659

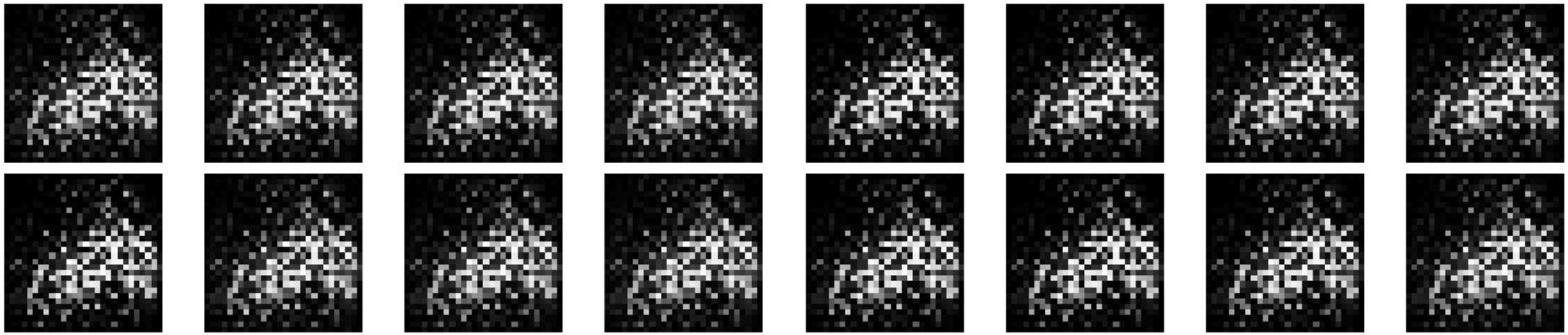
View run **vanilla_bce_adam** at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/rx4nvt47
View project at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: `./wandb/run-20260416_082151-rx4nvt47/logs`

```
starting: vanilla_bce_sgd
```
Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_082422-og156ku9`
Syncing run **vanilla_bce_sgd** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/og156ku9

```
  epoch 1/10  g_loss=0.7801  d_loss=0.3836
  epoch 2/10  g_loss=1.5528  d_loss=0.2888
  epoch 3/10  g_loss=2.4394  d_loss=0.1670
```

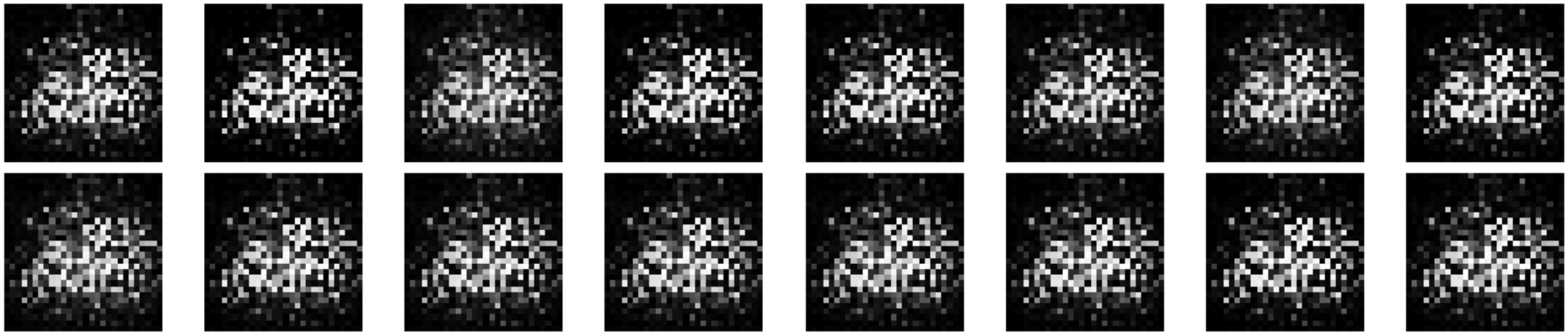
```
epoch 3/10  g_loss=2.4394  d_loss=0.1070
epoch 4/10  g_loss=2.8075  d_loss=0.1479
epoch 5/10  g_loss=3.0713  d_loss=0.1880
```

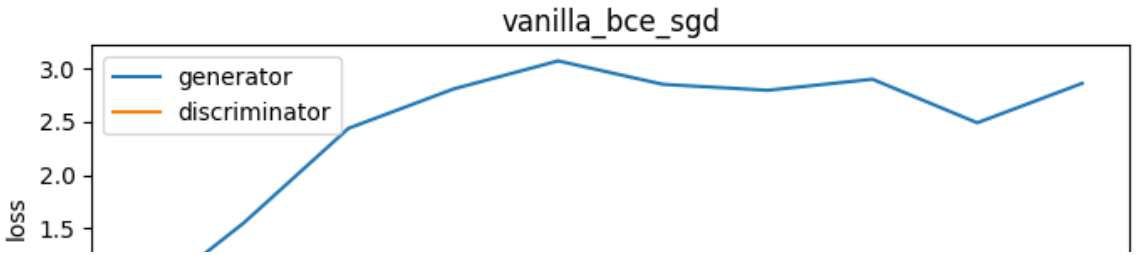
epoch 5 - vanilla_bce_sgd

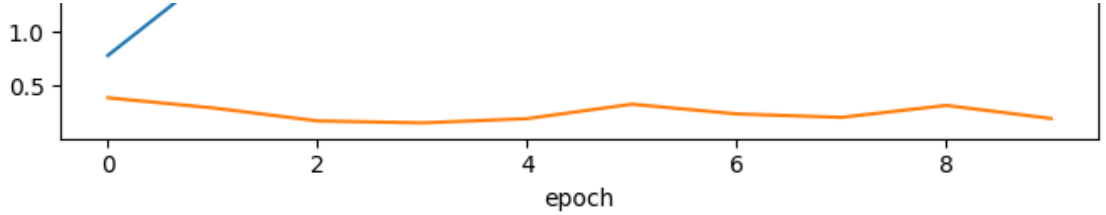


```
epoch 6/10  g_loss=2.8517  d_loss=0.3235
epoch 7/10  g_loss=2.7957  d_loss=0.2322
epoch 8/10  g_loss=2.8995  d_loss=0.1998
epoch 9/10  g_loss=2.4913  d_loss=0.3118
epoch 10/10  g_loss=2.8606  d_loss=0.1896
```

epoch 10 - vanilla_bce_sgd







Run history:

d_loss
epoch
g_loss

Run summary:

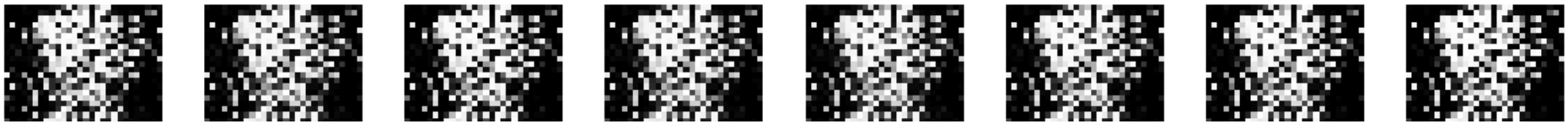
d_loss 0.18956
epoch 10
g_loss 2.86062

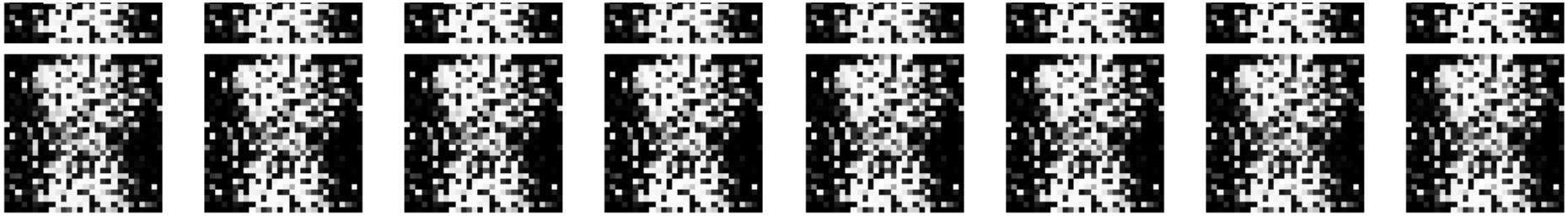
View run **vanilla_bce_sgd** at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/og156ku9
View project at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: `./wandb/run-20260416_082422-og156ku9/logs`

```
starting: vanilla_bce_rmsprop
```
Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_082703-dh2m2us9`
Syncing run **vanilla_bce_rmsprop** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/dh2m2us9

```
  epoch 1/10  g_loss=0.8590  d_loss=0.4701
  epoch 2/10  g_loss=1.3645  d_loss=0.3798
  epoch 3/10  g_loss=1.9841  d_loss=0.2958
  epoch 4/10  g_loss=2.3640  d_loss=0.2683
  epoch 5/10  g_loss=2.6548  d_loss=0.2546
```

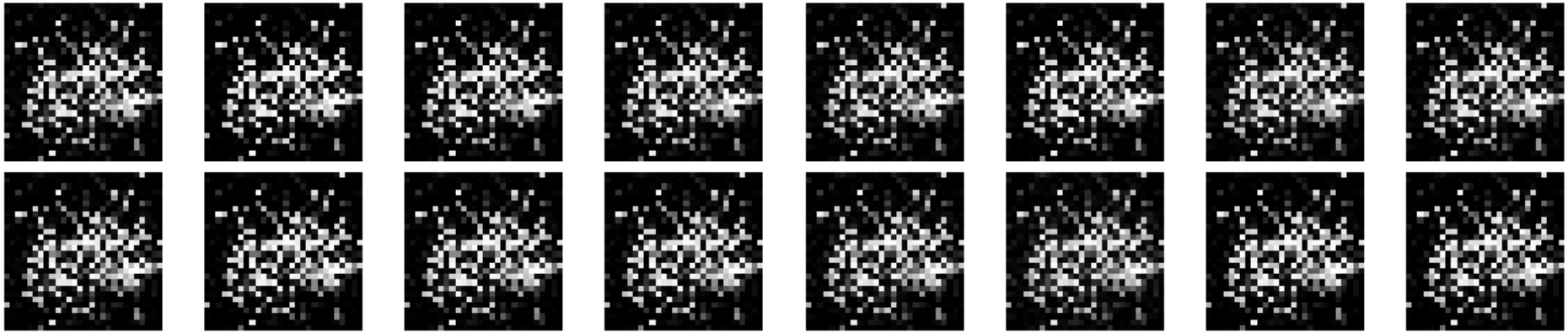
epoch 5 - vanilla_bce_rmsprop

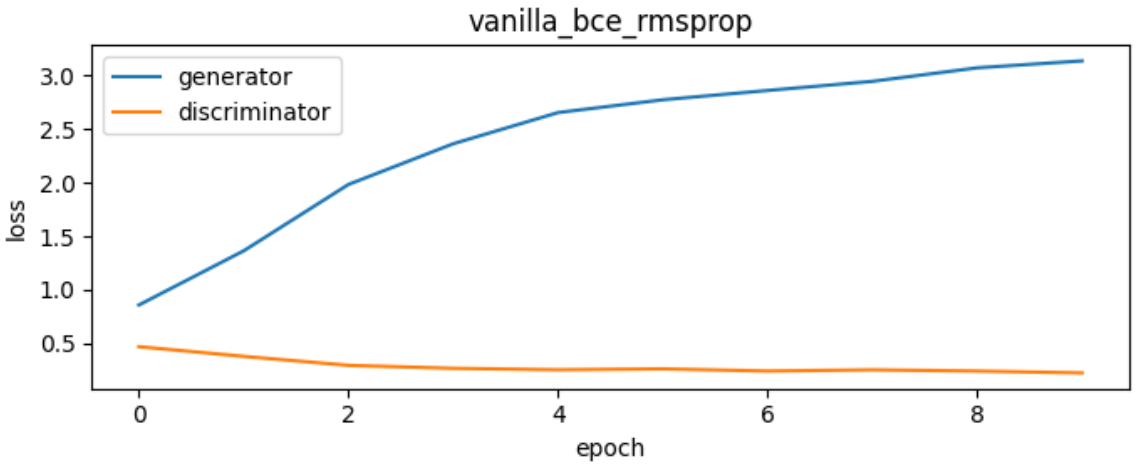




```
epoch 6/10  g_loss=2.7743  d_loss=0.2632
epoch 7/10  g_loss=2.8610  d_loss=0.2428
epoch 8/10  g_loss=2.9463  d_loss=0.2542
epoch 9/10  g_loss=3.0725  d_loss=0.2427
epoch 10/10  g_loss=3.1371  d_loss=0.2252
```

epoch 10 - vanilla_bce_rmsprop





Run history:

d_loss
epoch
g_loss

Run summary:

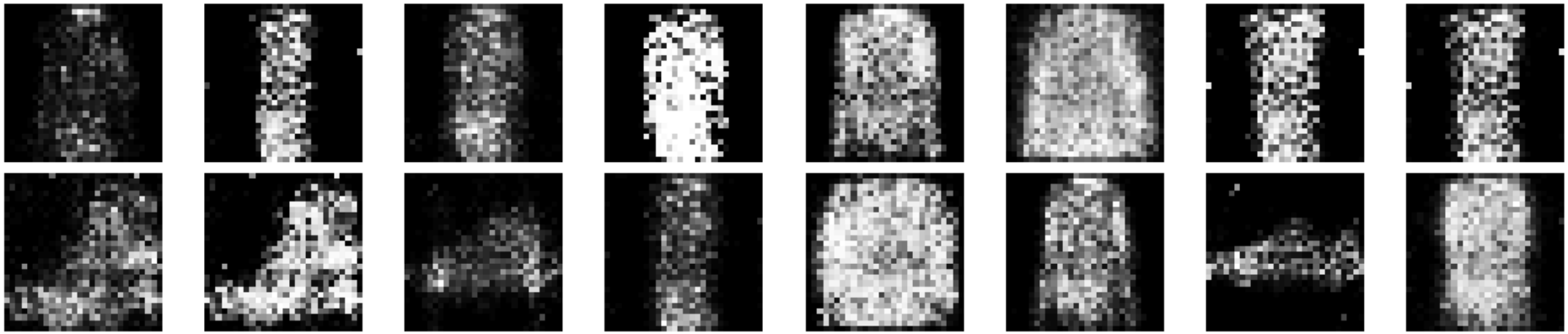
d_loss 0.22521
epoch 10
g_loss 3.13714

View run **vanilla_bce_rmsprop** at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/dh2m2us9
View project at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: `./wandb/run-20260416_082703-dh2m2us9/logs`

```
starting: vanilla_lsgan_adam
```
Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_082941-dqeb3idk`
Syncing run **vanilla_lsgan_adam** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/dqeb3idk

```
  epoch 1/10  g_loss=0.9906  d_loss=0.0717
  epoch 2/10  g_loss=0.8476  d_loss=0.1120
  epoch 3/10  g_loss=0.7473  d_loss=0.1272
  epoch 4/10  g_loss=0.6232  d_loss=0.1521
  epoch 5/10  g_loss=0.6011  d_loss=0.1626
```

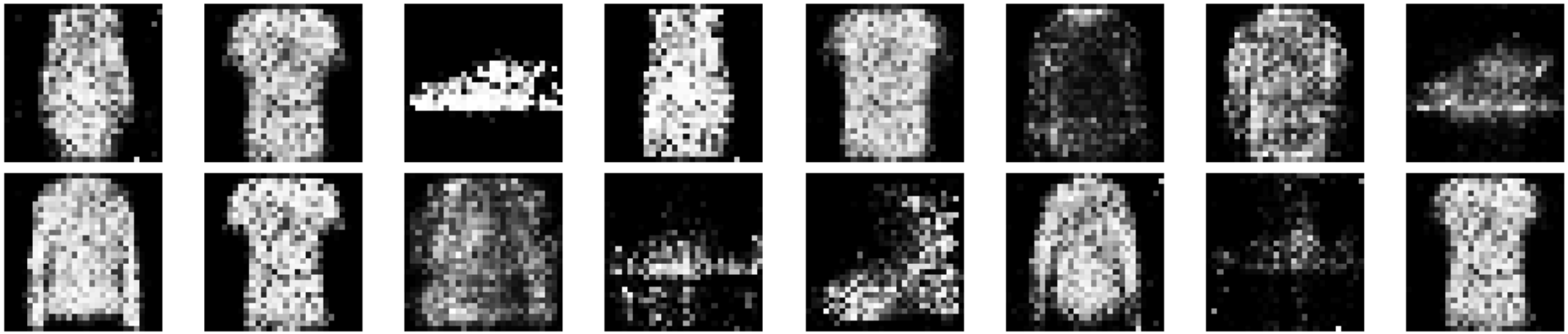
epoch 5 - vanilla_lsgan_adam

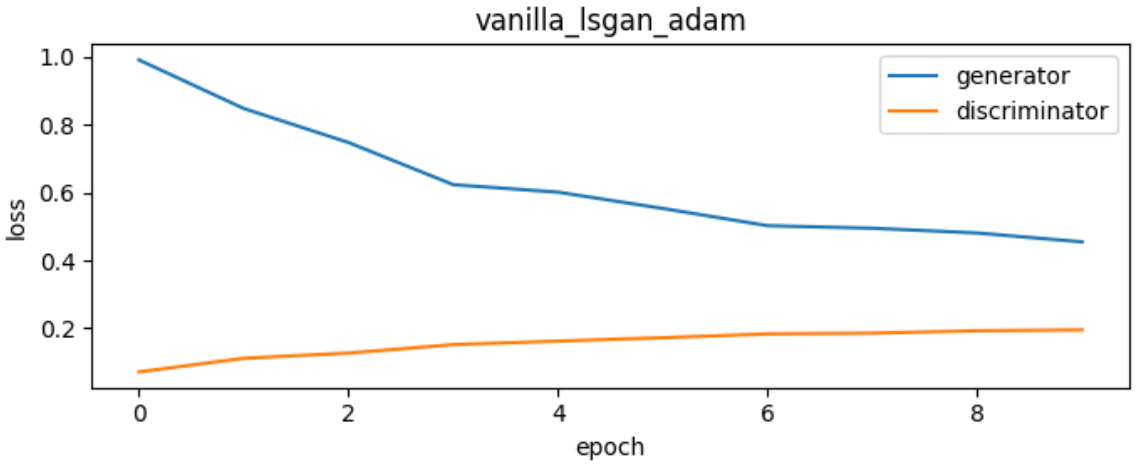


```
  epoch 6/10  g_loss=0.5531  d_loss=0.1725
```

```
epoch 7/10  g_loss=0.5024  d_loss=0.1836
epoch 8/10  g_loss=0.4949  d_loss=0.1860
epoch 9/10  g_loss=0.4809  d_loss=0.1930
epoch 10/10  g_loss=0.4547  d_loss=0.1957
```

epoch 10 - vanilla_lsgan_adam





Run history:

d_loss
epoch
g_loss

Run summary:

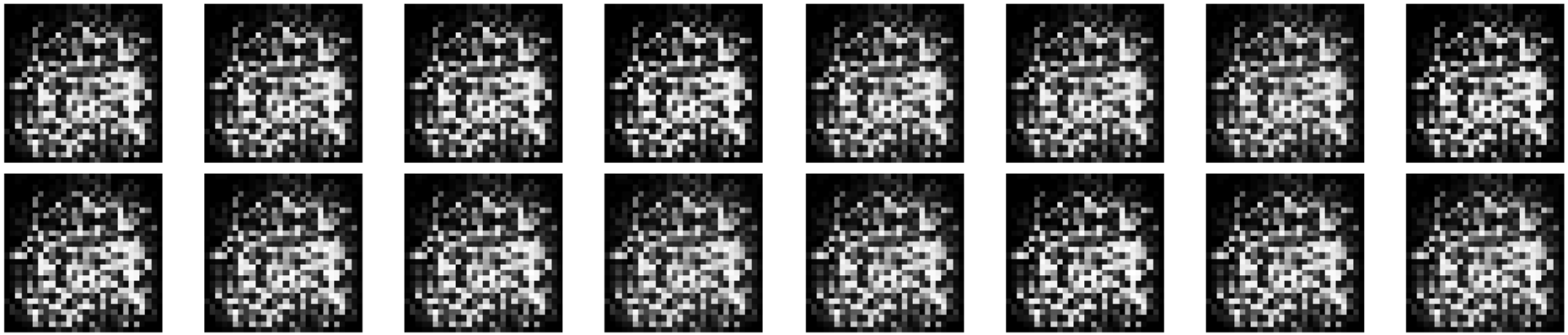
d_loss 0.19569
epoch 10
g_loss 0.45469

View run **vanilla_lsgan_adam** at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/dqeb3idk
View project at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: `./wandb/run-20260416_082941-dqeb3idk/logs`

```
starting: vanilla_lsgan_sgd
```
Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_083216-ouyh7m2r`
Syncing run **vanilla_lsgan_sgd** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/ouyh7m2r

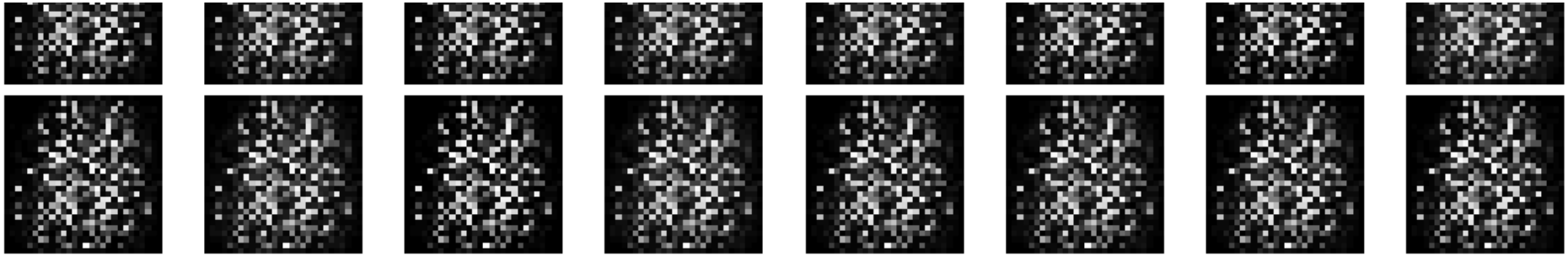
```
epoch 1/10  g_loss=0.7067  d_loss=0.0378
epoch 2/10  g_loss=0.9871  d_loss=0.0311
epoch 3/10  g_loss=0.9732  d_loss=0.0230
epoch 4/10  g_loss=0.9676  d_loss=0.0227
epoch 5/10  g_loss=0.9634  d_loss=0.0293
```

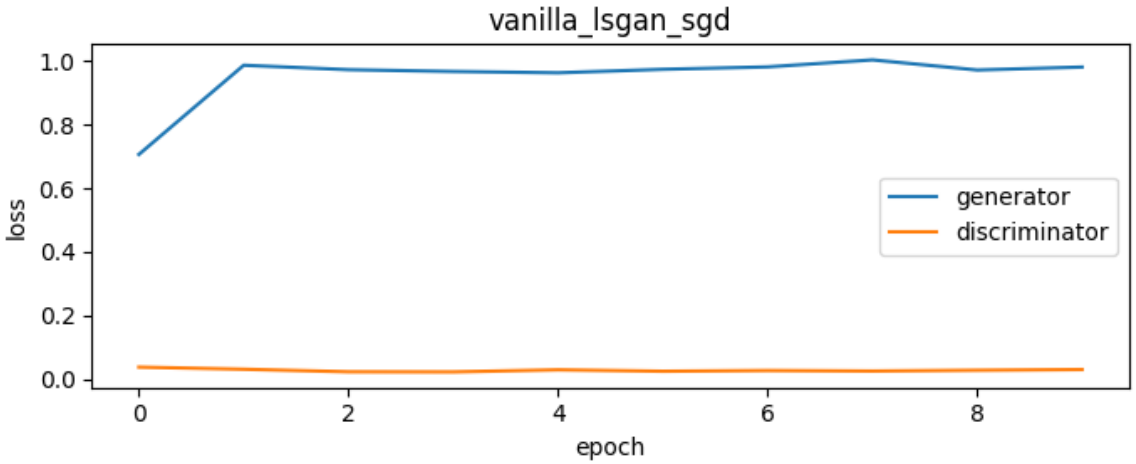
epoch 5 - vanilla_lsgan_sgd



```
epoch 6/10  g_loss=0.9744  d_loss=0.0248
epoch 7/10  g_loss=0.9819  d_loss=0.0270
epoch 8/10  g_loss=1.0040  d_loss=0.0254
epoch 9/10  g_loss=0.9722  d_loss=0.0280
epoch 10/10  g_loss=0.9815  d_loss=0.0302
```

epoch 10 - vanilla_lsgan_sgd





Run history:

d_loss [redacted]
epoch [redacted]
g_loss [redacted]

Run summary:

d_loss 0.03015
epoch 10
g_loss 0.98146

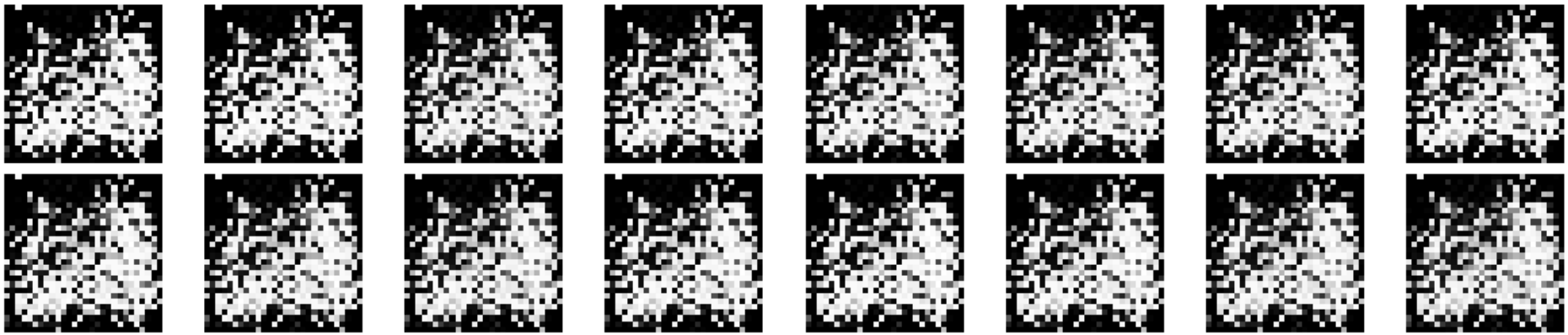
View run **vanilla_lsgan_sgd** at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/ouyh7m2r
View project at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: ./wandb/run-20260416_083216-ouyh7m2r/logs

```
starting: vanilla_lsgan_rmsprop
```

Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_083448-bsbhfoku`
Syncing run **vanilla_lsgan_rmsprop** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/bsbhfoku

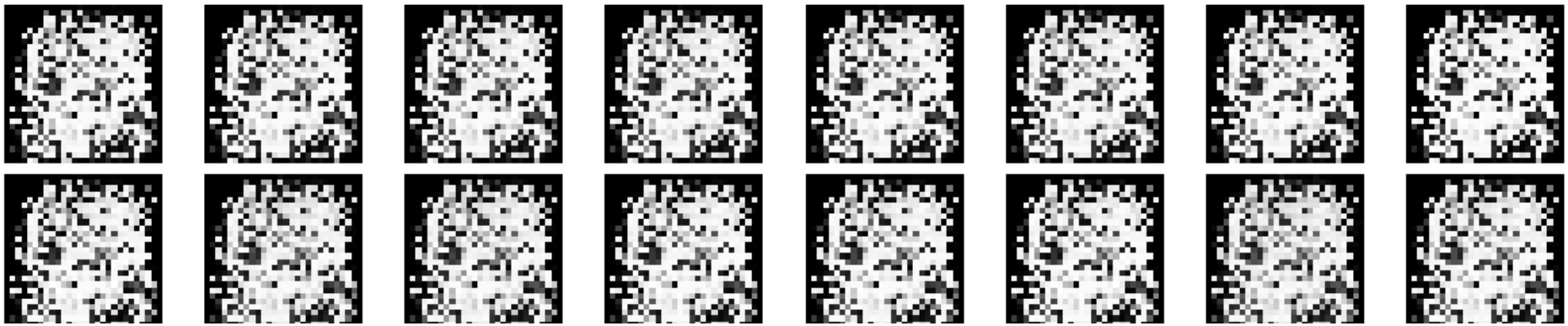
```
epoch 1/10  g_loss=0.8706  d_loss=0.0673
epoch 2/10  g_loss=0.9709  d_loss=0.0770
epoch 3/10  g_loss=0.9435  d_loss=0.0820
epoch 4/10  g_loss=0.9027  d_loss=0.0883
epoch 5/10  g_loss=0.9418  d_loss=0.0839
```

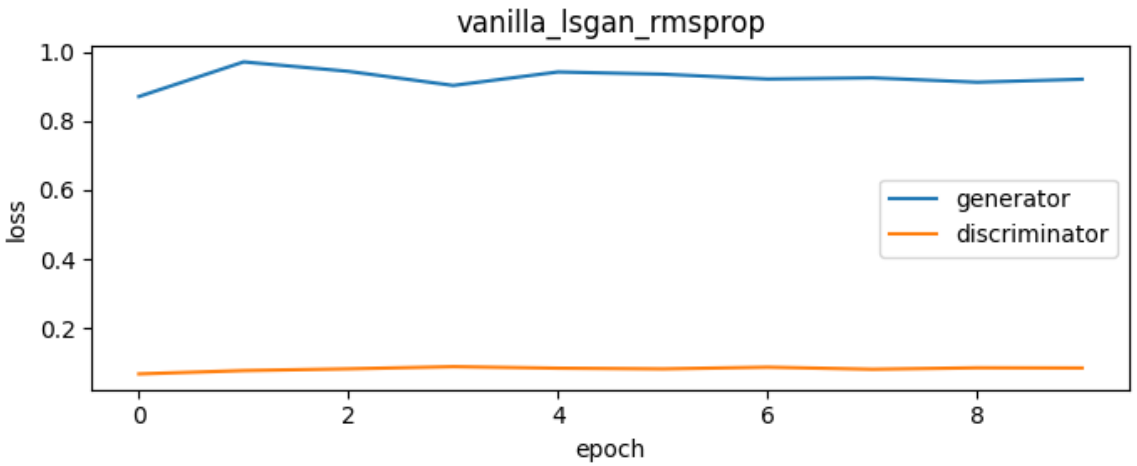
epoch 5 - vanilla_lsgan_rmsprop



```
epoch 6/10  g_loss=0.9353  d_loss=0.0818
epoch 7/10  g_loss=0.9213  d_loss=0.0869
epoch 8/10  g_loss=0.9247  d_loss=0.0808
epoch 9/10  g_loss=0.9121  d_loss=0.0850
epoch 10/10  g_loss=0.9206  d_loss=0.0844
```

epoch 10 - vanilla_lsgan_rmsprop





Run history:

d_loss
epoch
g_loss

Run summary:

d_loss 0.08444
epoch 10
g_loss 0.92061

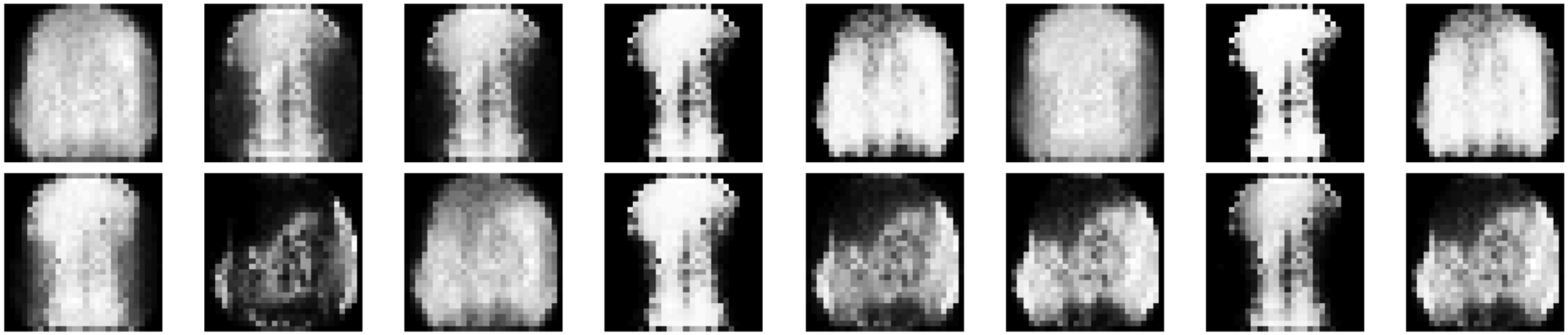
View run **vanilla_lsgan_rmsprop** at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/bsbhfoku
View project at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: `./wandb/run-20260416_083448-bsbhfoku/logs`

```
starting: vanilla_wgan_adam
```
Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_083718-y05e4rsk`
Syncing run **vanilla_wgan_adam** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/y05e4rsk

```
  epoch 1/10  g_loss=-0.3830  d_loss=-0.3851
  epoch 2/10  g_loss=-0.1386  d_loss=-0.0903
  epoch 3/10  g_loss=-0.0357  d_loss=-0.1950
```

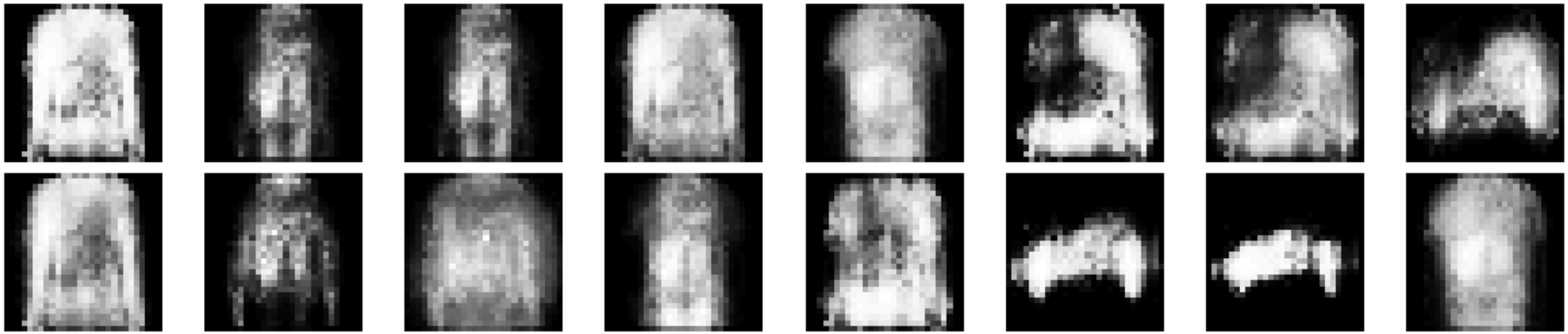
```
epoch 4/10  g_loss=-0.0592  d_loss=-0.0910
epoch 5/10  g_loss=-0.2461  d_loss=-0.1366
```

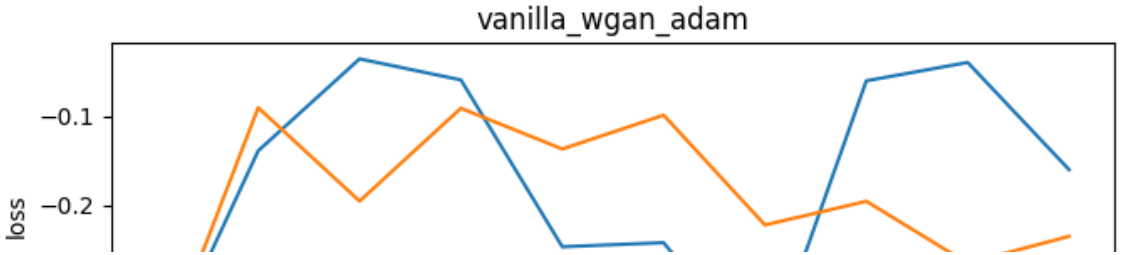
epoch 5 - vanilla_wgan_adam

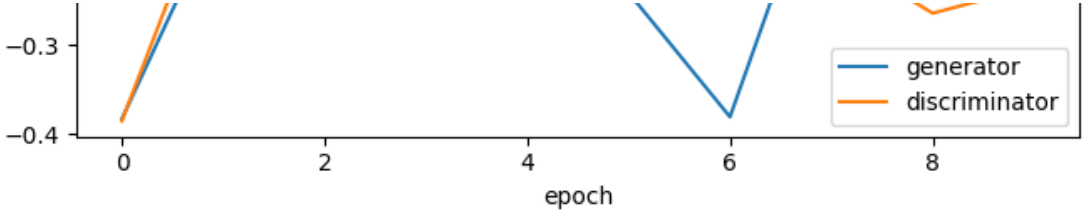


```
epoch 6/10  g_loss=-0.2415  d_loss=-0.0987
epoch 7/10  g_loss=-0.3806  d_loss=-0.2217
epoch 8/10  g_loss=-0.0603  d_loss=-0.1952
epoch 9/10  g_loss=-0.0397  d_loss=-0.2645
epoch 10/10  g_loss=-0.1597  d_loss=-0.2345
```

epoch 10 - vanilla_wgan_adam







Run history:

d_loss
epoch
g_loss

Run summary:

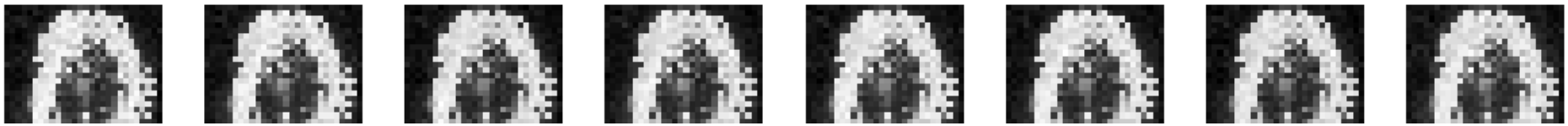
d_loss -0.23445
epoch 10
g_loss -0.15972

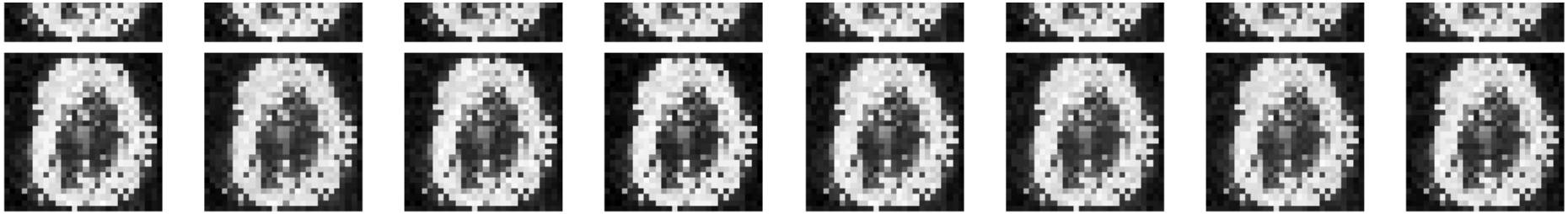
View run **vanilla_wgan_adam** at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/y05e4rsk
View project at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: `./wandb/run-20260416_083718-y05e4rsk/logs`

```
starting: vanilla_wgan_sgd
```
Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_083953-pmxh0jez`
Syncing run **vanilla_wgan_sgd** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/pmxh0jez

```
  epoch 1/10  g_loss=0.1740  d_loss=-1.8285
  epoch 2/10  g_loss=2.1473  d_loss=-0.6807
  epoch 3/10  g_loss=2.7006  d_loss=-0.8318
  epoch 4/10  g_loss=-0.8761  d_loss=-3.0942
  epoch 5/10  g_loss=3.1445  d_loss=-1.0674
```

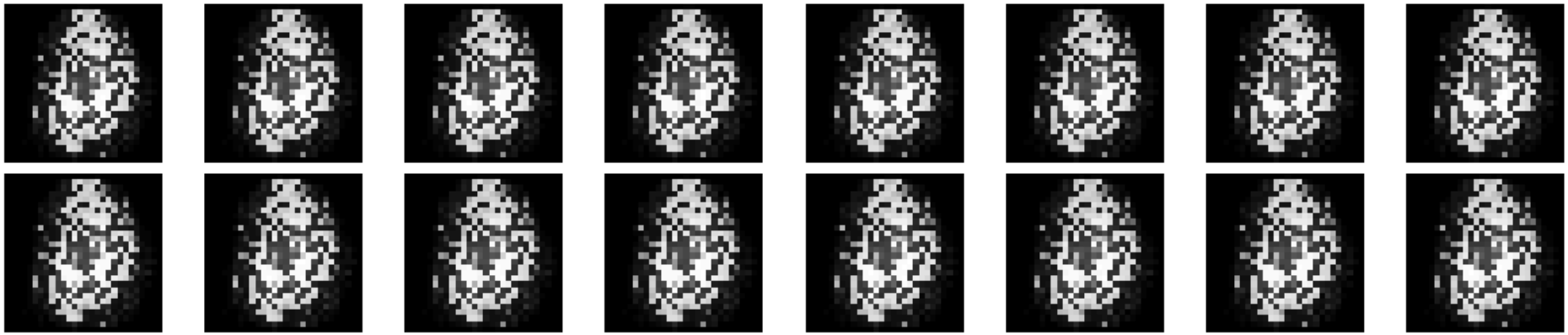
epoch 5 - vanilla_wgan_sgd

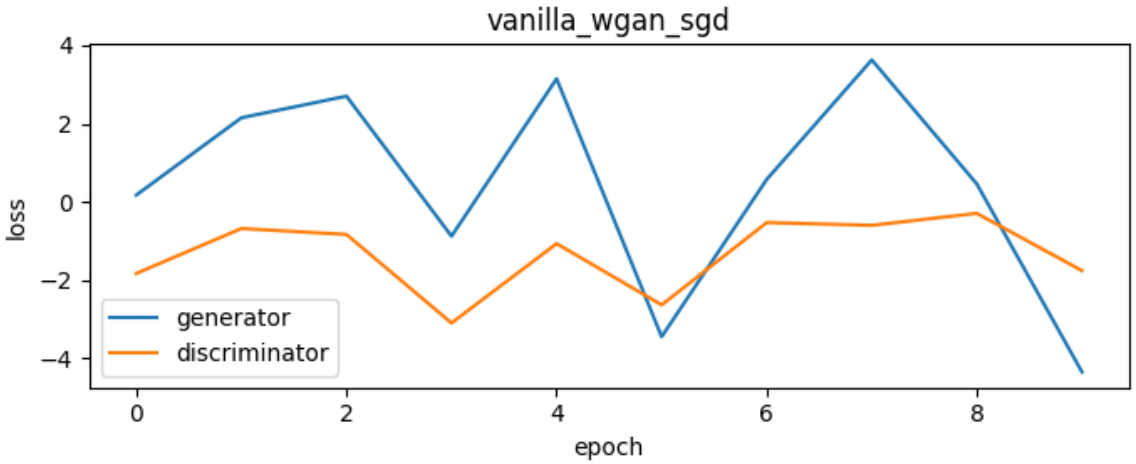




```
epoch 6/10  g_loss=-3.4453  d_loss=-2.6321
epoch 7/10  g_loss=0.5732  d_loss=-0.5288
epoch 8/10  g_loss=3.6268  d_loss=-0.5995
epoch 9/10  g_loss=0.4555  d_loss=-0.2944
epoch 10/10  g_loss=-4.3443  d_loss=-1.7566
```

epoch 10 - vanilla_wgan_sgd





Run history:

d_loss
epoch
g_loss

Run summary:

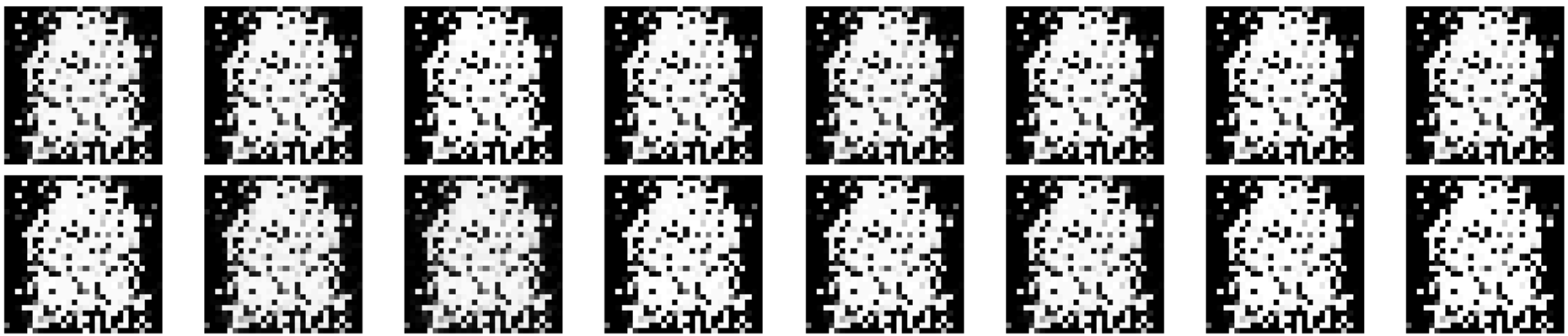
d_loss -1.75665
epoch 10
g_loss -4.34428

View run **vanilla_wgan_sgd** at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/pmxh0jez
View project at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: `./wandb/run-20260416_083953-pmxh0jez/logs`

```
starting: vanilla_wgan_rmsprop
```
Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_084230-hhht5vfy`
Syncing run **vanilla_wgan_rmsprop** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/hhht5vfy

```
epoch 1/10  g_loss=-0.0707  d_loss=-0.2090
epoch 2/10  g_loss=-0.1364  d_loss=-0.2003
epoch 3/10  g_loss=-0.3603  d_loss=-0.2775
epoch 4/10  g_loss=-0.5149  d_loss=-0.2504
epoch 5/10  g_loss=-0.1688  d_loss=-0.5607
```

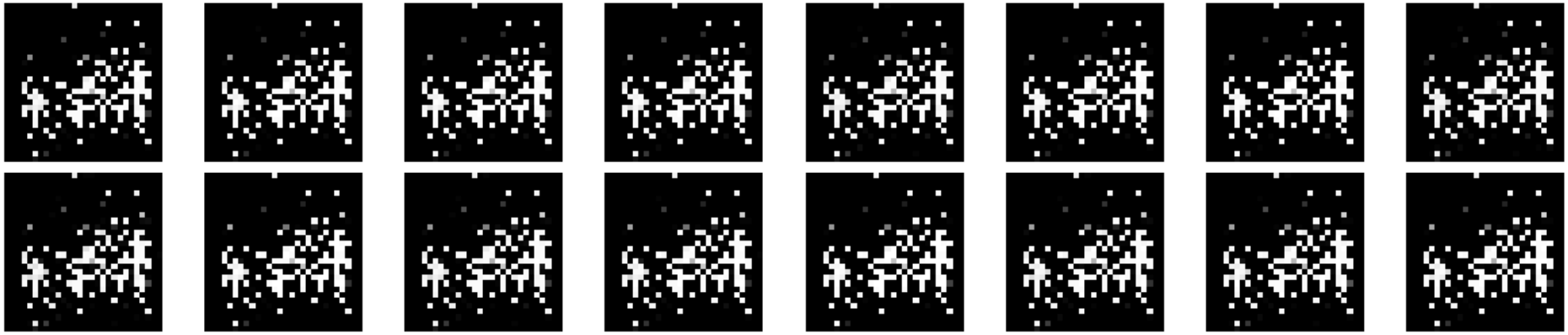
epoch 5 - vanilla_wgan_rmsprop

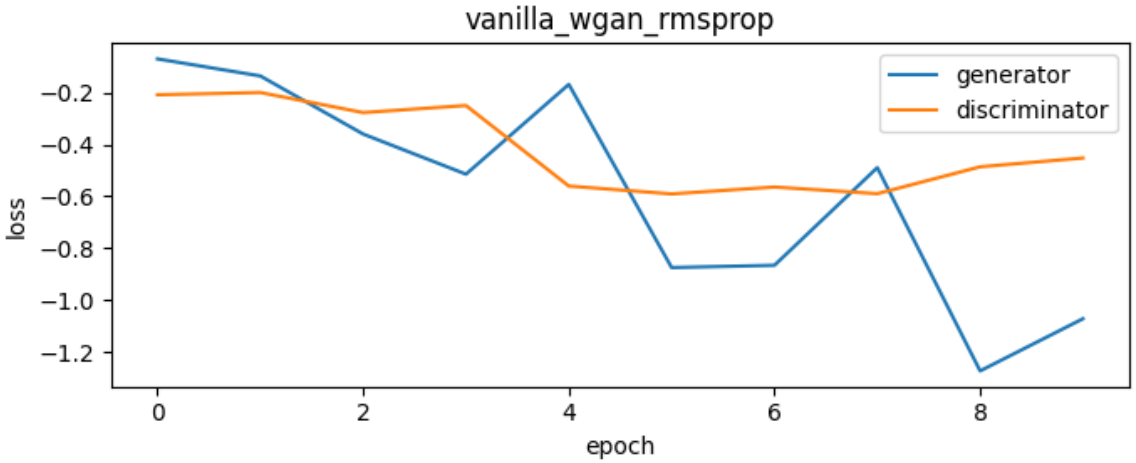


```
epoch 6/10  g_loss=-0.8748  d_loss=-0.5911
```

```
epoch 7/10  g_loss=-0.8661  d_loss=-0.5645
epoch 8/10  g_loss=-0.4896  d_loss=-0.5901
epoch 9/10  g_loss=-1.2732  d_loss=-0.4865
epoch 10/10  g_loss=-1.0721  d_loss=-0.4527
```

epoch 10 - vanilla_wgan_rmsprop





Run history:

d_loss
epoch
g_loss

Run summary:

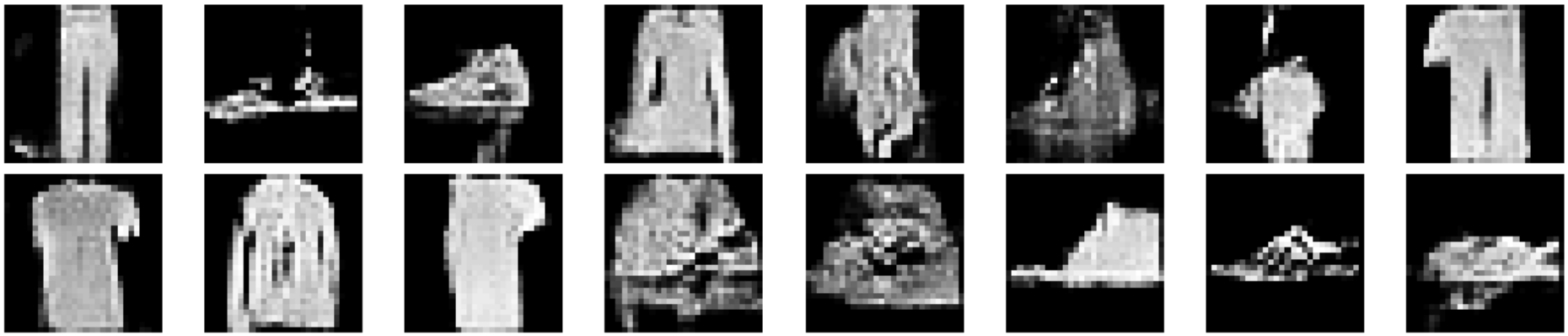
d_loss -0.45267
epoch 10
g_loss -1.07206

View run **vanilla_wgan_rmsprop** at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/hhht5vfy
View project at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: `./wandb/run-20260416_084230-hhht5vfy/logs`

```
starting: dcgan_bce_adam
```
Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_084504-pwfk9v7s`
Syncing run **dcgan_bce_adam** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/pwfk9v7s

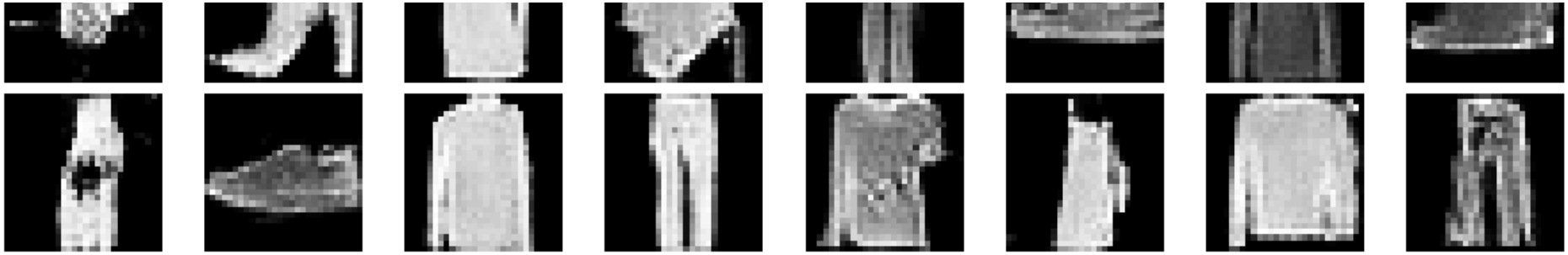
```
epoch 1/10  g_loss=1.4030  d_loss=0.4084
epoch 2/10  g_loss=1.2523  d_loss=0.4573
epoch 3/10  g_loss=1.0955  d_loss=0.5277
epoch 4/10  g_loss=1.0638  d_loss=0.5430
epoch 5/10  g_loss=1.0672  d_loss=0.5418
```

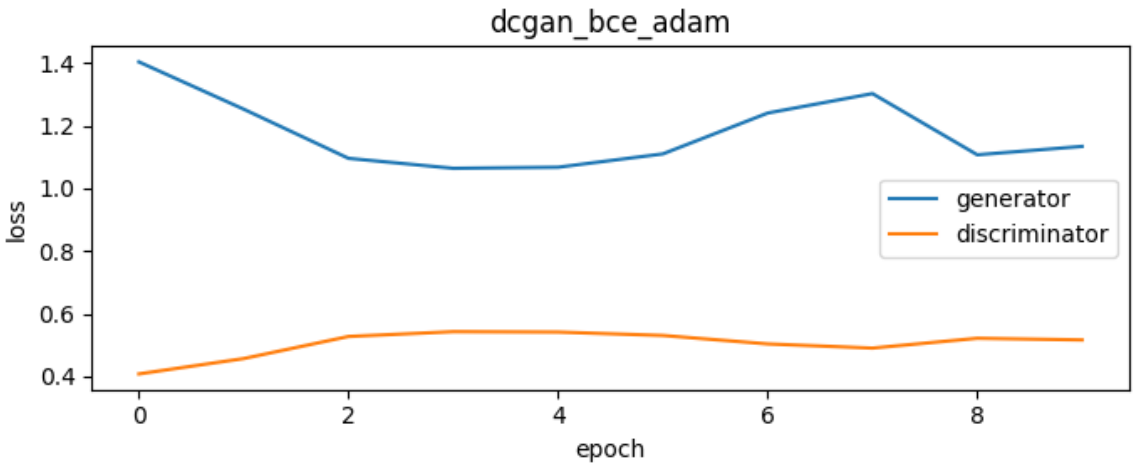
epoch 5 - dcgan_bce_adam



```
epoch 6/10  g_loss=1.1097  d_loss=0.5309
epoch 7/10  g_loss=1.2399  d_loss=0.5040
epoch 8/10  g_loss=1.3019  d_loss=0.4907
epoch 9/10  g_loss=1.1069  d_loss=0.5220
epoch 10/10  g_loss=1.1336  d_loss=0.5168
```

epoch 10 - dcgan_bce_adam





Run history:

d_loss [redacted]
epoch [redacted]
g_loss [redacted]

Run summary:

d_loss 0.51678
epoch 10
g_loss 1.1336

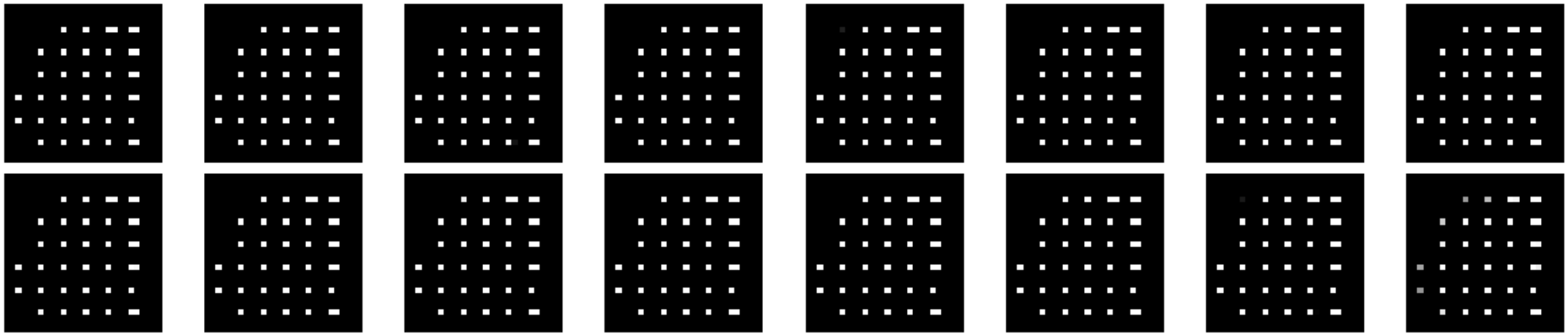
View run **dcgan_bce_adam** at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/pwfk9v7s
View project at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: ./wandb/run-20260416_084504-pwfk9v7s/logs

```
starting: dcgan_bce_sgd
```

Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_084806-s44ofa66`
Syncing run **dcgan_bce_sgd** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/s44ofa66

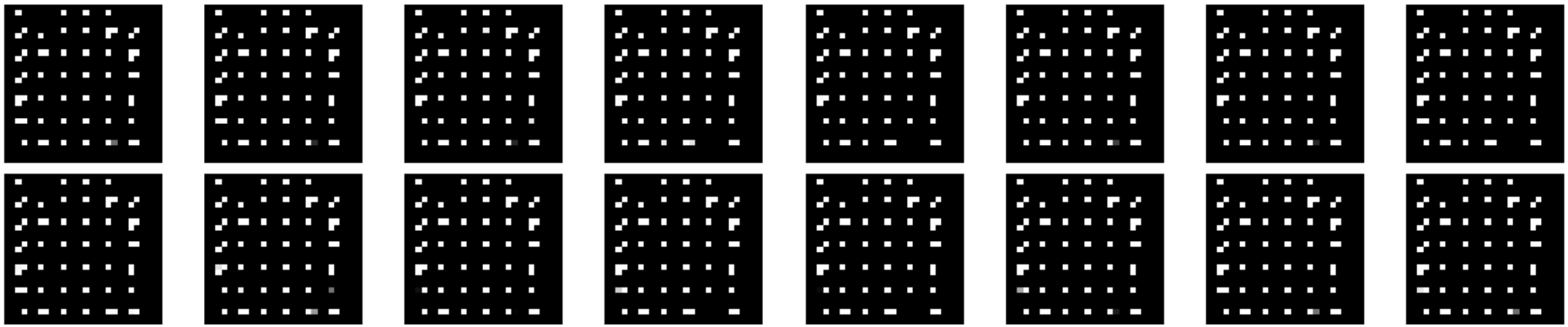
```
epoch 1/10  g_loss=7.2758  d_loss=0.0258
epoch 2/10  g_loss=8.2248  d_loss=0.0015
epoch 3/10  g_loss=11.4000  d_loss=0.0025
epoch 4/10  g_loss=10.0358  d_loss=0.0003
epoch 5/10  g_loss=9.8548  d_loss=0.0002
```

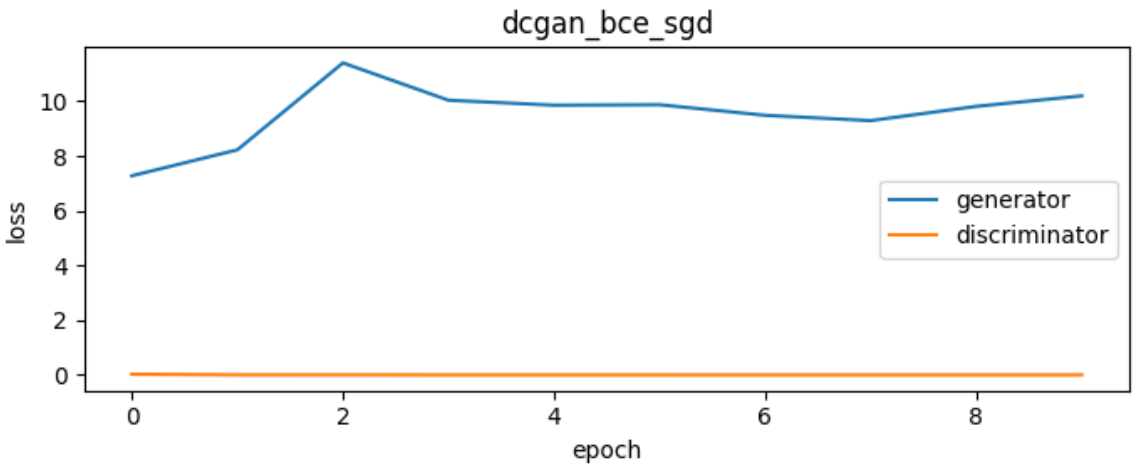
epoch 5 - dcgan_bce_sgd



```
epoch 6/10  g_loss=9.8716  d_loss=0.0001
epoch 7/10  g_loss=9.4856  d_loss=0.0001
epoch 8/10  g_loss=9.2904  d_loss=0.0001
epoch 9/10  g_loss=9.8120  d_loss=0.0001
epoch 10/10  g_loss=10.1939  d_loss=0.0001
```

epoch 10 - dcgan_bce_sgd





Run history:

d_loss
epoch
g_loss

Run summary:

d_loss 9e-05
epoch 10
g_loss 10.19394

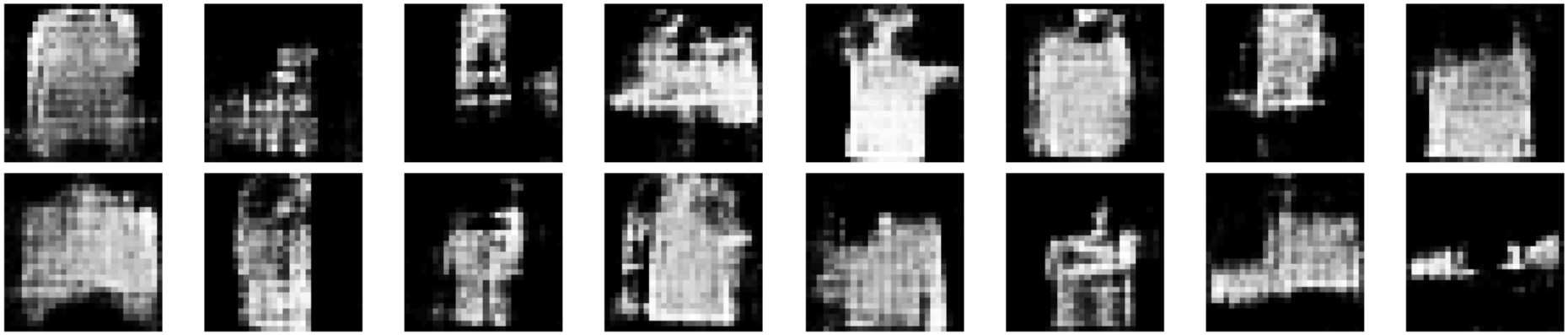
View run **dcgan_bce_sgd** at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/s44ofa66
View project at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: `./wandb/run-20260416_084806-s44ofa66/logs`

```
starting: dcgan_bce_rmsprop
```
Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_085102-jcqe6397`
Syncing run **dcgan_bce_rmsprop** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/jcqe6397

```
epoch 1/10  g_loss=4.8516  d_loss=0.0336
epoch 2/10  g_loss=7.1295  d_loss=0.0477
epoch 3/10  g_loss=1.1843  d_loss=0.4411
```

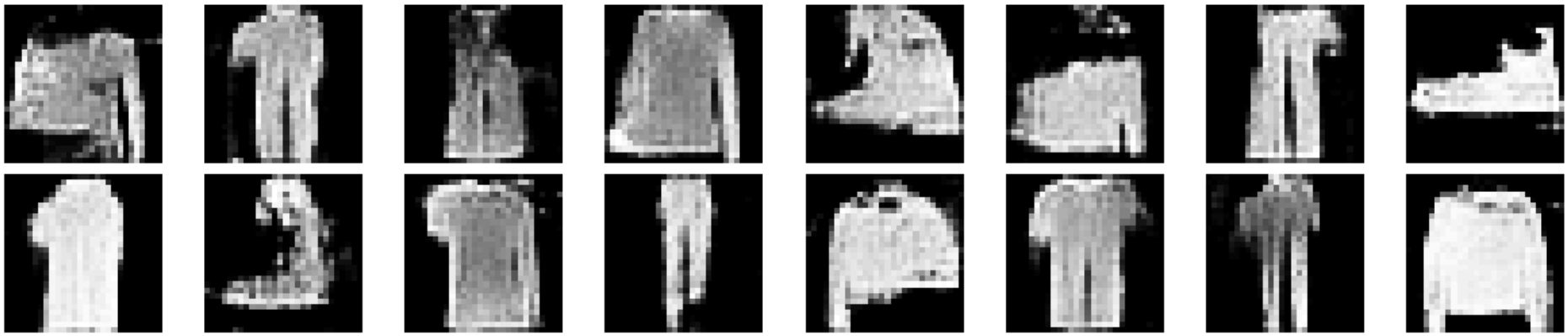
```
epoch 4/10  g_loss=1.5607  d_loss=0.3013
epoch 5/10  g_loss=1.3894  d_loss=0.3641
```

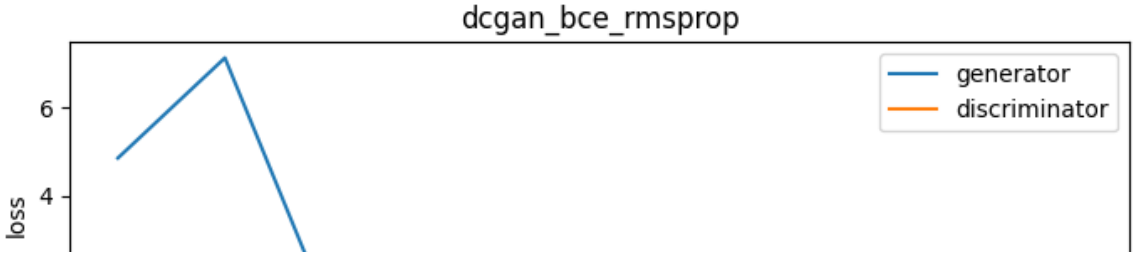
epoch 5 - dcgan_bce_rmsprop

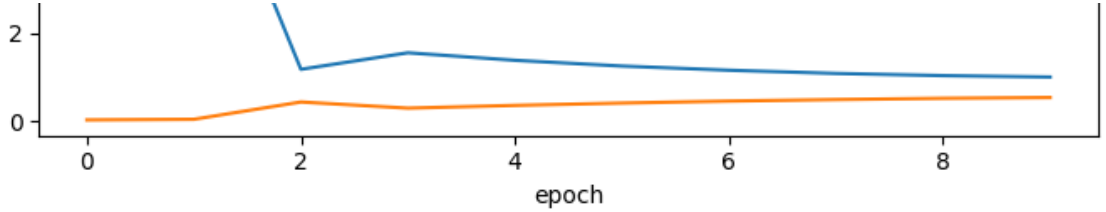


```
epoch 6/10  g_loss=1.2585  d_loss=0.4171
epoch 7/10  g_loss=1.1636  d_loss=0.4624
epoch 8/10  g_loss=1.0908  d_loss=0.4965
epoch 9/10  g_loss=1.0413  d_loss=0.5252
epoch 10/10  g_loss=1.0089  d_loss=0.5426
```

epoch 10 - dcgan_bce_rmsprop







Run history:

d_loss
epoch
g_loss

Run summary:

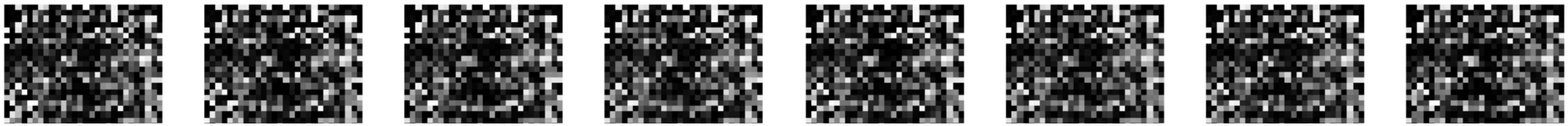
d_loss 0.54256
epoch 10
g_loss 1.00886

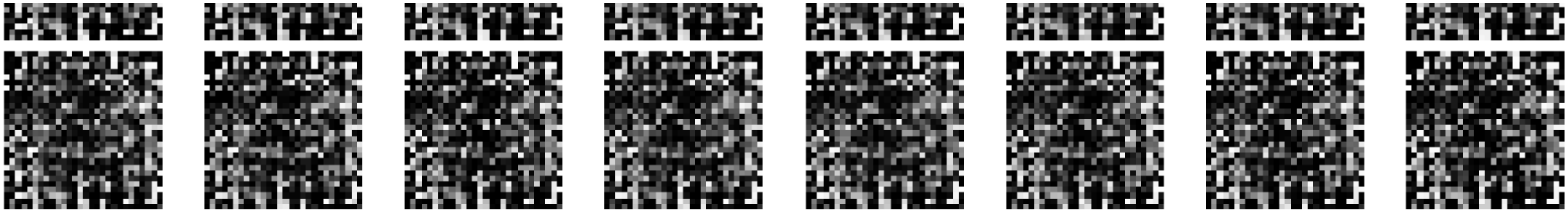
View run **dcgan_bce_rmsprop** at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/jcqe6397
View project at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: `./wandb/run-20260416_085102-jcqe6397/logs`

```
starting: dcgan_lsgan_adam
```
Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_085359-kixitms7`
Syncing run **dcgan_lsgan_adam** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/kixitms7

```
  epoch 1/10  g_loss=1.0690  d_loss=0.0425
  epoch 2/10  g_loss=1.0398  d_loss=0.0236
  epoch 3/10  g_loss=1.0644  d_loss=0.0307
  epoch 4/10  g_loss=1.1569  d_loss=0.0463
  epoch 5/10  g_loss=1.0817  d_loss=0.0345
```

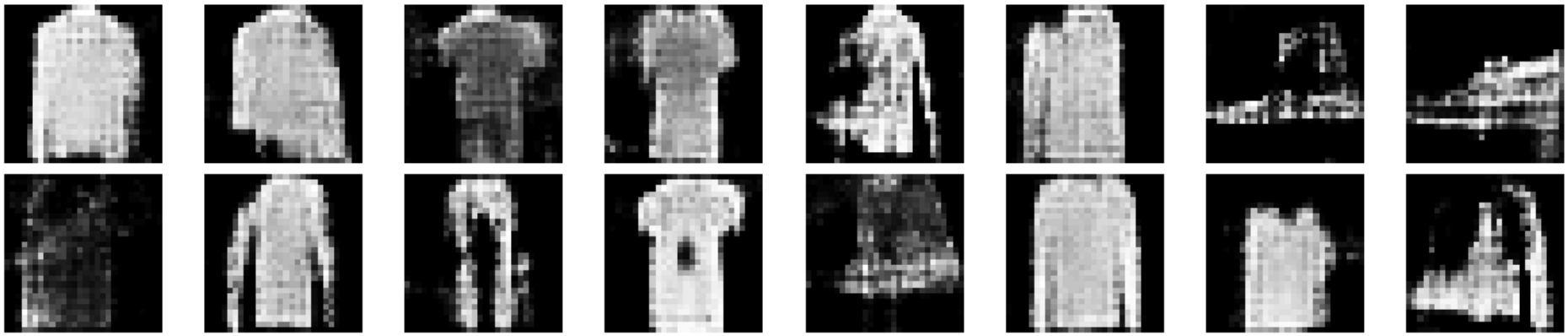
epoch 5 - dcgan_lsgan_adam

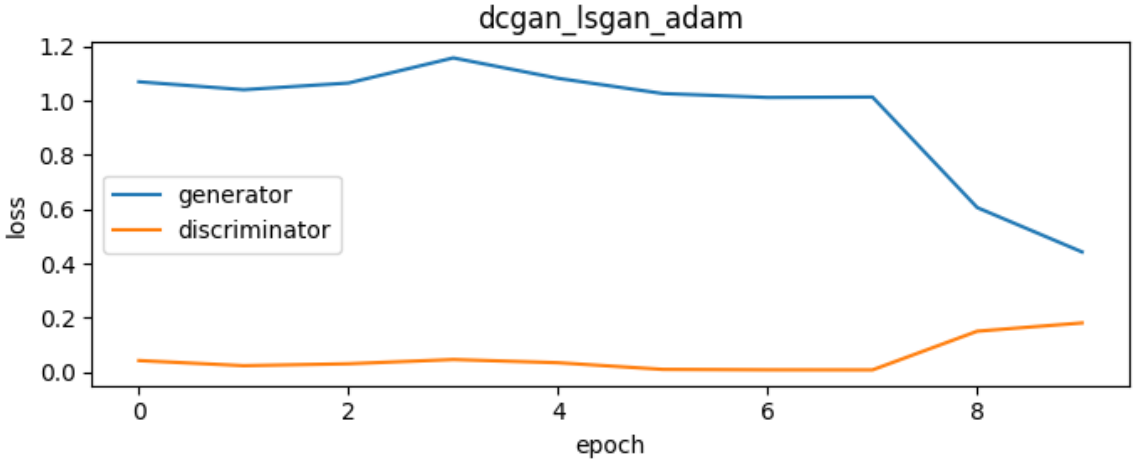




```
epoch 6/10  g_loss=1.0256  d_loss=0.0100
epoch 7/10  g_loss=1.0113  d_loss=0.0083
epoch 8/10  g_loss=1.0128  d_loss=0.0080
epoch 9/10  g_loss=0.6057  d_loss=0.1506
epoch 10/10  g_loss=0.4424  d_loss=0.1810
```

epoch 10 - dcgan_lsgan_adam





Run history:

d_loss
epoch
g_loss

Run summary:

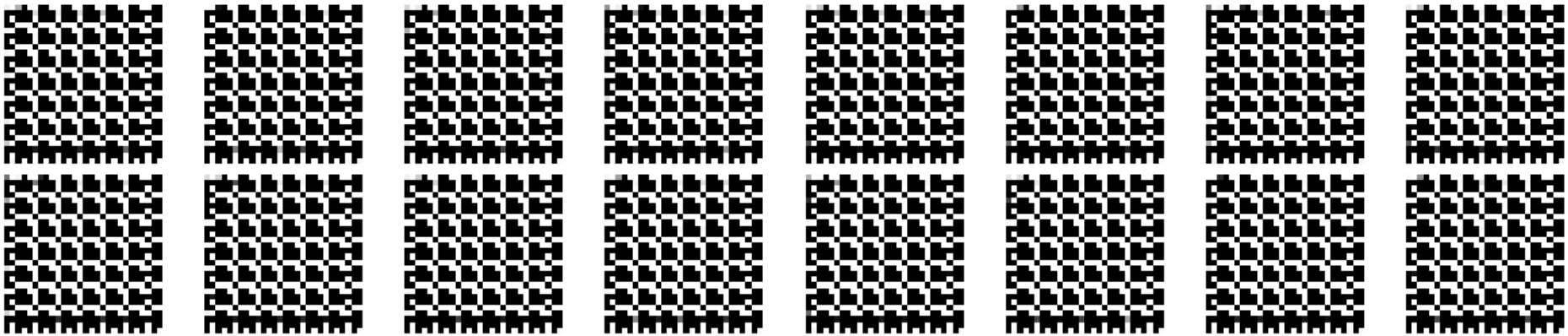
d_loss 0.18102
epoch 10
g_loss 0.44238

View run **dcgan_lsgan_adam** at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/kixitms7
View project at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: `./wandb/run-20260416_085359-kixitms7/logs`

```
starting: dcgan_lsgan_sgd
```
Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_085655-nuix5kz7`
Syncing run **dcgan_lsgan_sgd** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/nuix5kz7

```
epoch 1/10  g_loss=15.0390  d_loss=8.3897
epoch 2/10  g_loss=0.9944  d_loss=0.0053
epoch 3/10  g_loss=0.9972  d_loss=0.0030
epoch 4/10  g_loss=0.9976  d_loss=0.0022
epoch 5/10  g_loss=0.9985  d_loss=0.0018
```

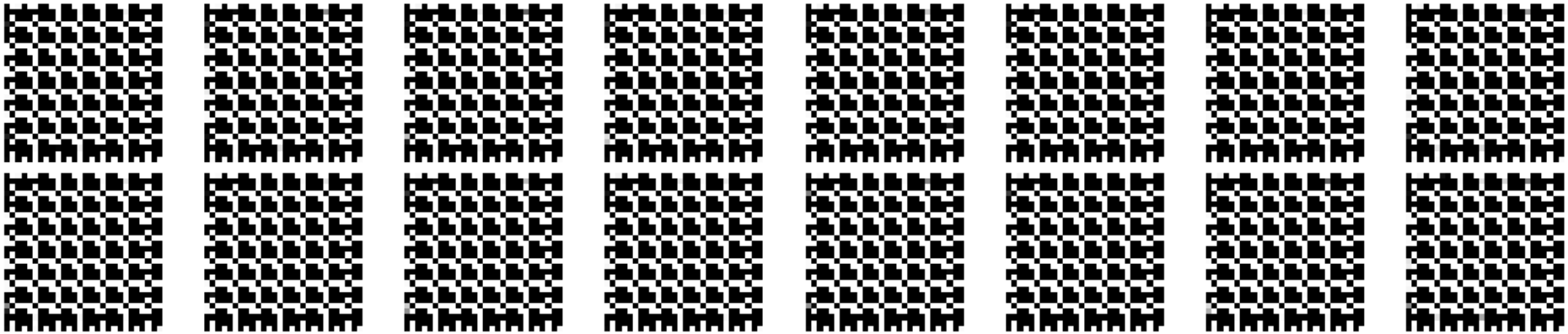
epoch 5 - dcgan_lsgan_sgd

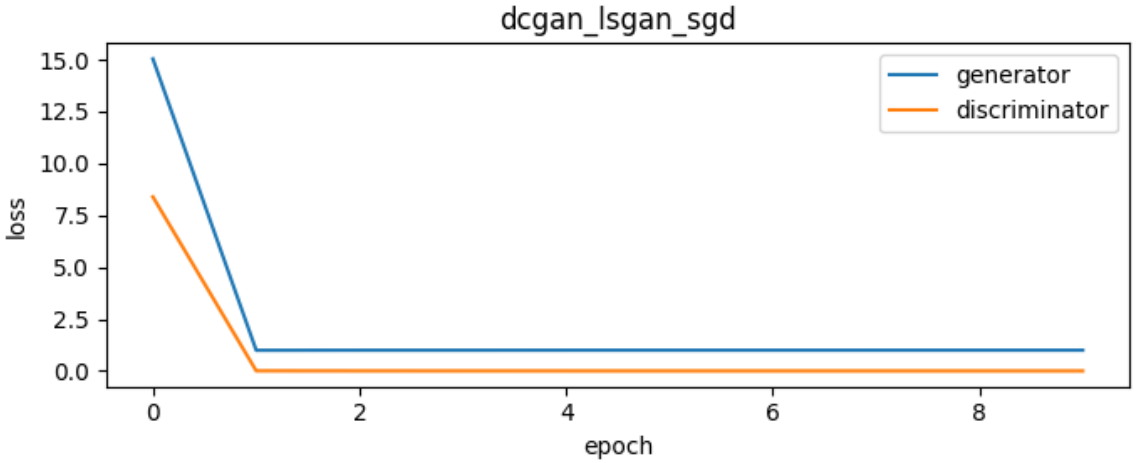


```
epoch 6/10  g_loss=0.9987  d_loss=0.0015
```

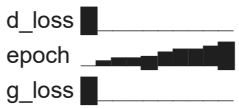
```
epoch 7/10  g_loss=0.9991  d_loss=0.0013
epoch 8/10  g_loss=0.9990  d_loss=0.0012
epoch 9/10  g_loss=0.9995  d_loss=0.0011
epoch 10/10  g_loss=0.9991  d_loss=0.0010
```

epoch 10 - dcgan_lsgan_sgd





Run history:



Run summary:

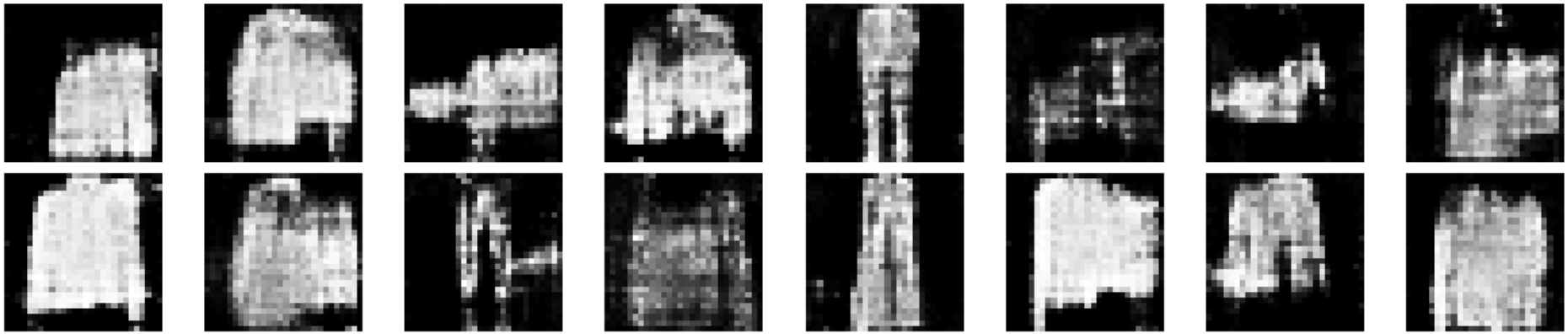
d_loss 0.00098
epoch 10
g_loss 0.99913

View run **dcgan_lsgan_sgd** at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/nuix5kz7
View project at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: `./wandb/run-20260416_085655-nuix5kz7/logs`

```
starting: dcgan_lsgan_rmsprop
```
Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_085951-xrhebhui`
Syncing run **dcgan_lsgan_rmsprop** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/xrhebhui

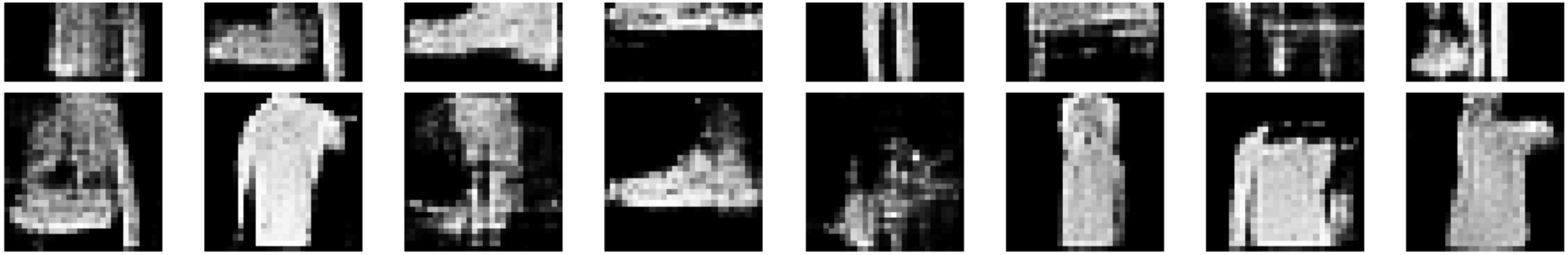
```
epoch 1/10  g_loss=1.0373  d_loss=0.0430
epoch 2/10  g_loss=1.0202  d_loss=0.0210
epoch 3/10  g_loss=0.5322  d_loss=0.1697
epoch 4/10  g_loss=0.5751  d_loss=0.1170
epoch 5/10  g_loss=0.5796  d_loss=0.1142
```

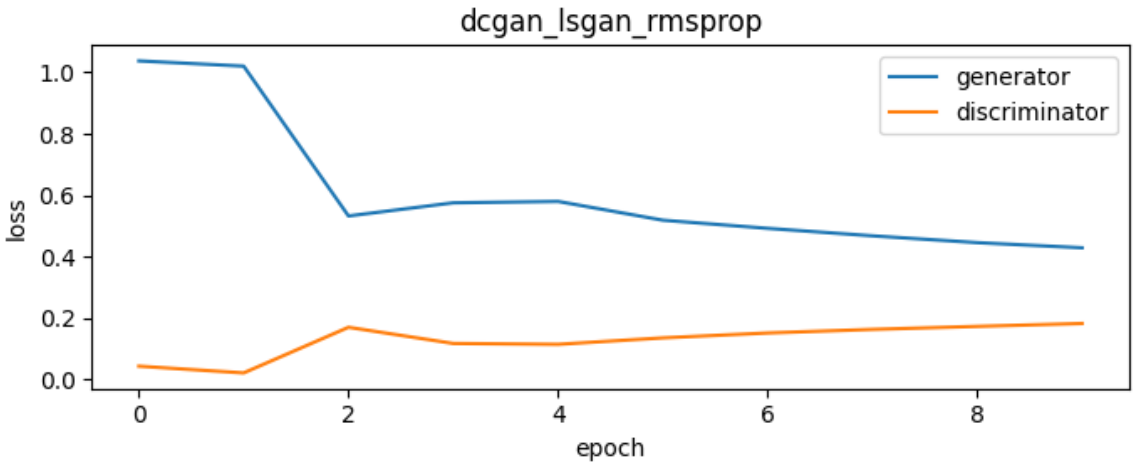
epoch 5 - dcgan_lsgan_rmsprop



```
epoch 6/10  g_loss=0.5184  d_loss=0.1352
epoch 7/10  g_loss=0.4919  d_loss=0.1511
epoch 8/10  g_loss=0.4675  d_loss=0.1628
epoch 9/10  g_loss=0.4453  d_loss=0.1726
epoch 10/10  g_loss=0.4286  d_loss=0.1820
```

epoch 10 - dcgan_lsgan_rmsprop





Run history:

d_loss
epoch
g_loss

Run summary:

d_loss 0.18201
epoch 10
g_loss 0.42857

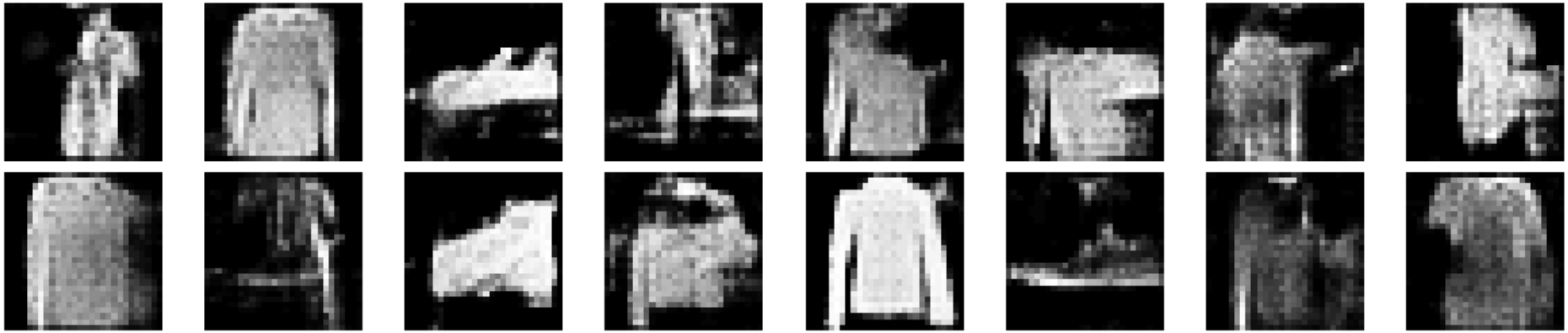
View run **dcgan_lsgan_rmsprop** at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/xrhebhui
View project at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: ./wandb/run-20260416_085951-xrhebhui/logs

```
starting: dcgan_wgan_adam
```

Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_090246-q6i1cwu3`
Syncing run **dcgan_wgan_adam** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/q6i1cwu3

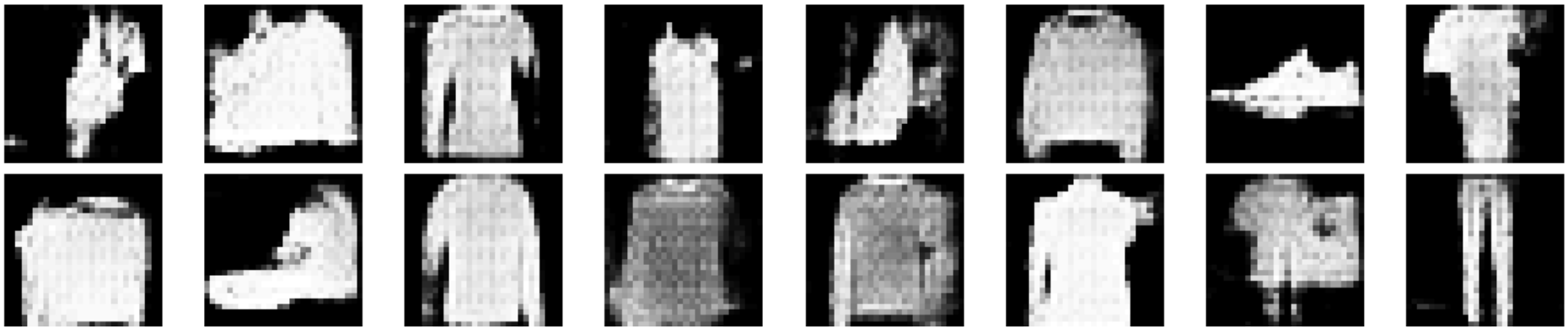
```
epoch 1/10  g_loss=0.0037  d_loss=-0.0218
epoch 2/10  g_loss=0.0128  d_loss=-0.0108
epoch 3/10  g_loss=0.0039  d_loss=-0.0094
epoch 4/10  g_loss=0.0051  d_loss=-0.0071
epoch 5/10  g_loss=-0.0013  d_loss=-0.0053
```

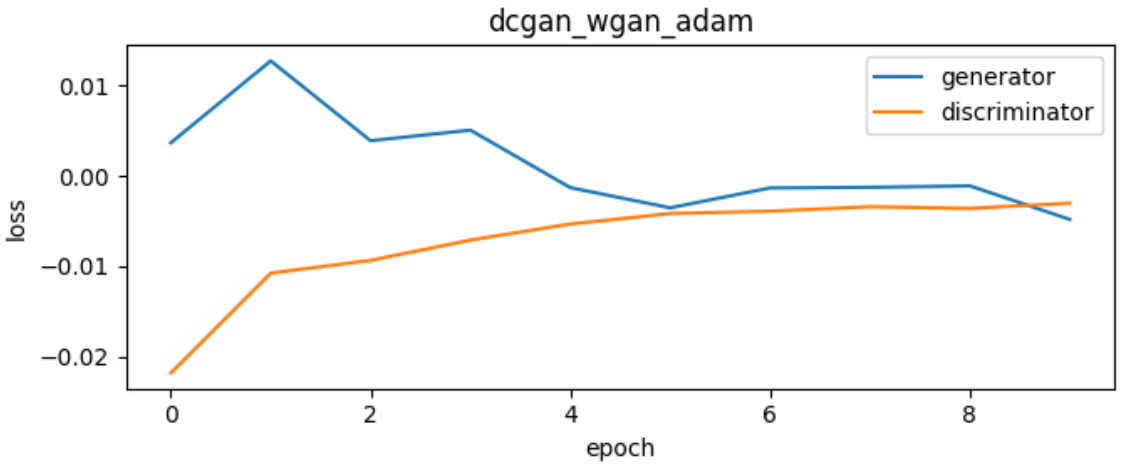
epoch 5 - dcgan_wgan_adam



```
epoch 6/10  g_loss=-0.0035  d_loss=-0.0042
epoch 7/10  g_loss=-0.0013  d_loss=-0.0039
epoch 8/10  g_loss=-0.0013  d_loss=-0.0034
epoch 9/10  g_loss=-0.0011  d_loss=-0.0036
epoch 10/10  g_loss=-0.0048  d_loss=-0.0030
```

epoch 10 - dcgan_wgan_adam





Run history:

d_loss [redacted]
epoch [redacted]
g_loss [redacted]

Run summary:

d_loss -0.00303
epoch 10
g_loss -0.00482

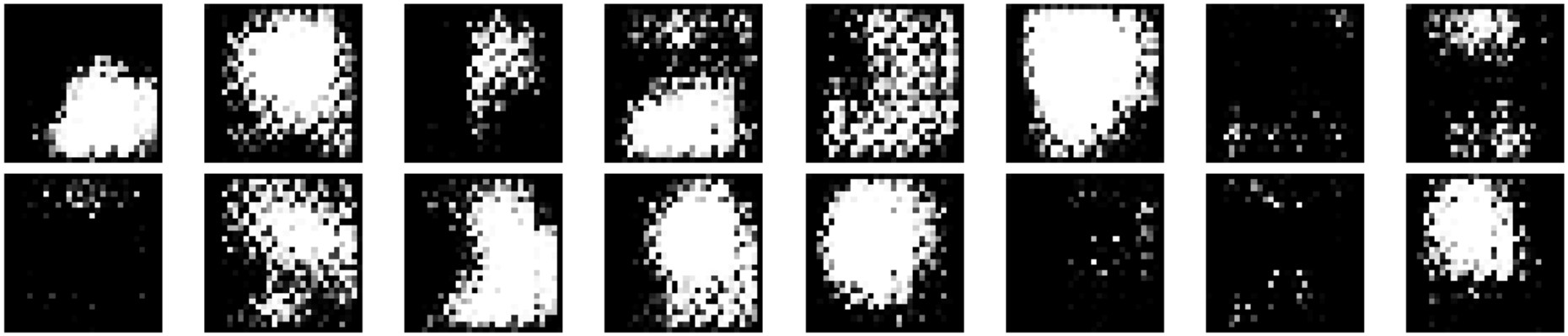
View run **dcgan_wgan_adam** at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/q6i1cwu3
View project at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: `./wandb/run-20260416_090246-q6i1cwu3/logs`

```
starting: dcgan_wgan_sgd
```
Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_090543-tvsgupfc`
Syncing run **dcgan_wgan_sgd** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/tvsgupfc

```
epoch 1/10  g_loss=0.2397  d_loss=-0.4436
epoch 2/10  g_loss=0.4491  d_loss=-0.8043
epoch 3/10  g_loss=0.4494  d_loss=-0.8347
```

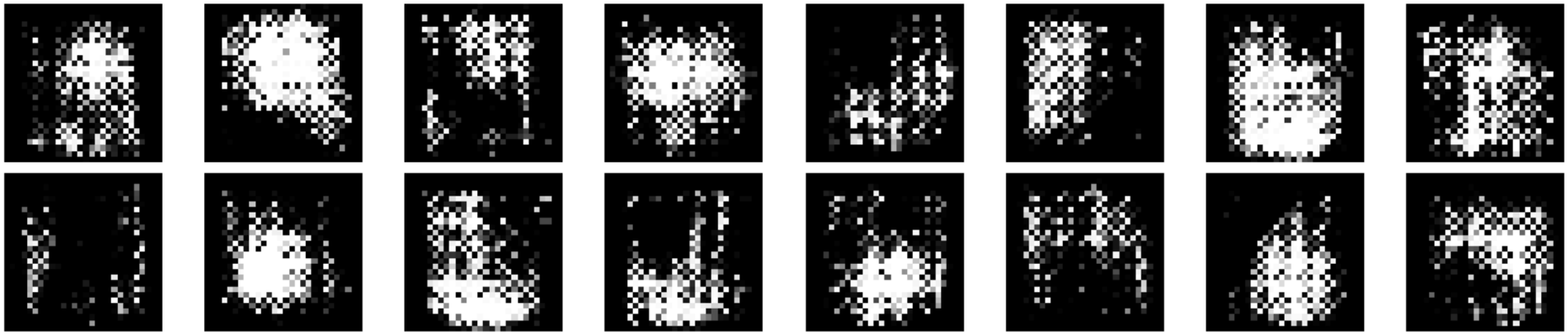
```
epoch 4/10  g_loss=0.3857  d_loss=-0.7354
epoch 5/10  g_loss=0.1797  d_loss=-0.4421
```

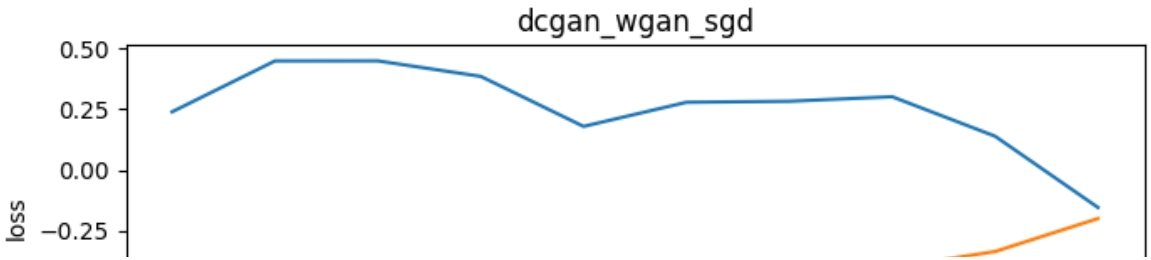
epoch 5 - dcgan_wgan_sgd

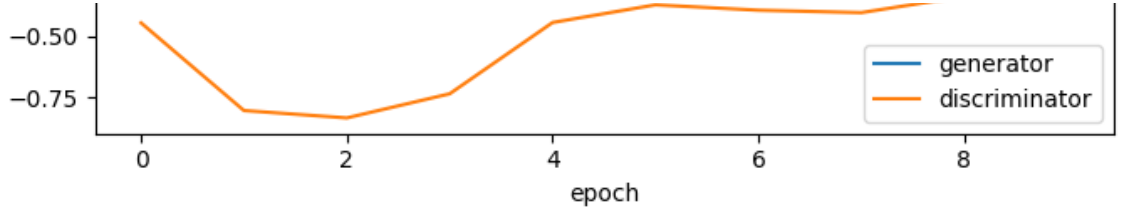


```
epoch 6/10  g_loss=0.2790  d_loss=-0.3693
epoch 7/10  g_loss=0.2833  d_loss=-0.3914
epoch 8/10  g_loss=0.3019  d_loss=-0.4015
epoch 9/10  g_loss=0.1394  d_loss=-0.3352
epoch 10/10  g_loss=-0.1540  d_loss=-0.1994
```

epoch 10 - dcgan_wgan_sgd







Run history:

d_loss
epoch
g_loss

Run summary:

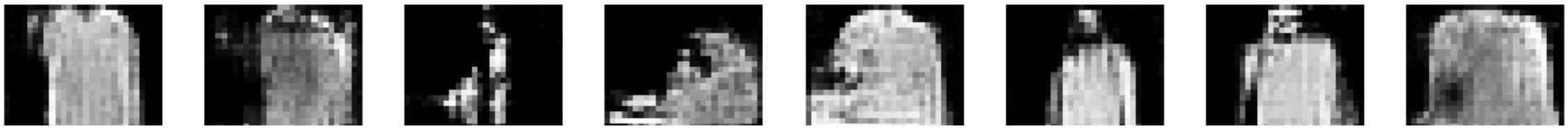
d_loss -0.19941
epoch 10
g_loss -0.15396

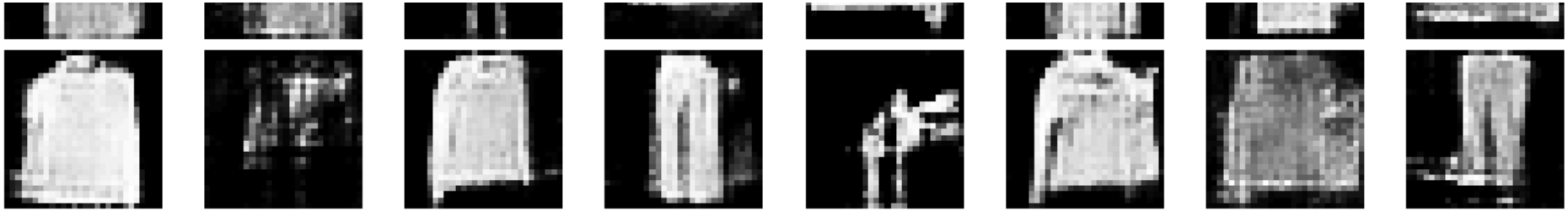
View run **dcgan_wgan_sgd** at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/tvsgupfc
View project at: https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 3 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: `./wandb/run-20260416_090543-tvsgupfc/logs`

```
starting: dcgan_wgan_rmsprop
```
Tracking run with wandb version 0.25.1
Run data is saved locally in `/content/wandb/run-20260416_090836-vtdel48n`
Syncing run **dcgan_wgan_rmsprop** to Weights & Biases (docs)
View project at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans
View run at https://wandb.ai/mayankjain_25rco06-delhi-technological-university/exp9-gans/runs/vtdel48n

```
  epoch 1/10  g_loss=0.0036  d_loss=-0.0305
  epoch 2/10  g_loss=0.0134  d_loss=-0.0111
  epoch 3/10  g_loss=0.0208  d_loss=-0.0246
  epoch 4/10  g_loss=0.0239  d_loss=-0.0219
  epoch 5/10  g_loss=0.0242  d_loss=-0.0172
```

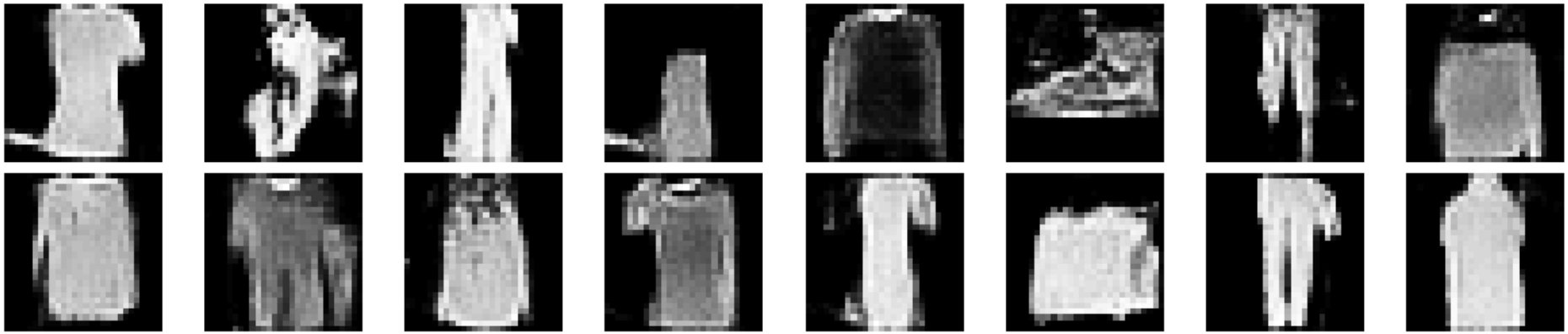
epoch 5 - dcgan_wgan_rmsprop

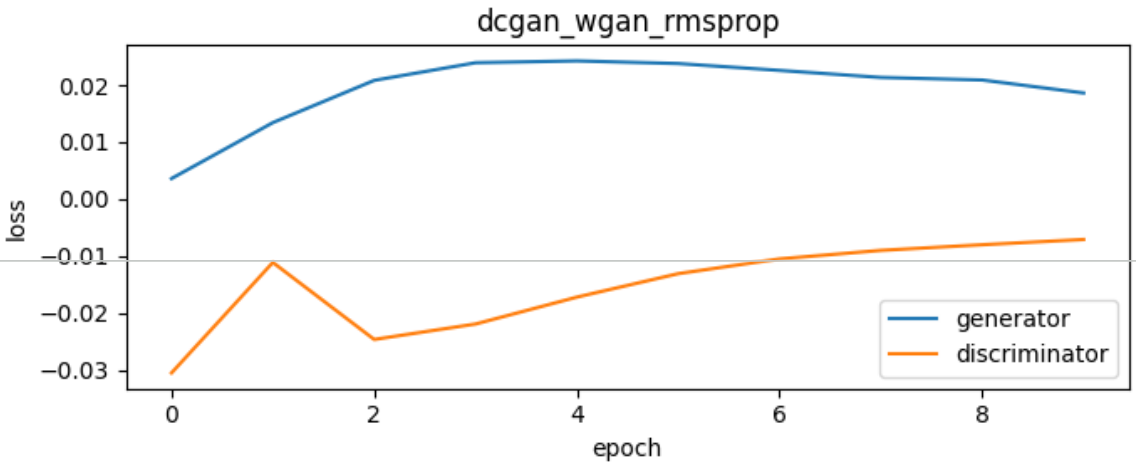




```
epoch 6/10  g_loss=0.0238  d_loss=-0.0131
epoch 7/10  g_loss=0.0226  d_loss=-0.0105
epoch 8/10  g_loss=0.0213  d_loss=-0.0090
epoch 9/10  g_loss=0.0209  d_loss=-0.0080
epoch 10/10  g_loss=0.0186  d_loss=-0.0071
```

epoch 10 - dcgan_wgan_rmsprop





Results Summary

Run history:

d_loss [redacted]

```
print(f'{"config":<35} {"g_loss":>10} {"d_loss":>10}')
print('-' * 57)
for k, v in sorted(results.items(), key=lambda x: x[1]['g_loss']):
    print(f'{k:<35} {v["g_loss"]:>10.4f} {v["d_loss"]:>10.4f}')
```

Run summary:

d_loss -0.00708
epoch 10
g_loss 0.01361

```
config                                  g_loss     d_loss
---------------------------------------------------------
vanilla_wgan_sgd                       -4.3443    -1.7566
vanilla_wgan_rmsprop                   -1.0721    -0.4527
vanilla_wgan_adam                      -0.1597    -0.2345
dcgan_wgan_sgd                         -0.1540    -0.1994
dcgan_wgan_adam                        -0.0048    -0.0030
dcgan_wgan_rmsprop                      0.0186     0.0071
dcgan_lsgan_rmsprop                     0.4286     0.1820
dcgan_lsgan_adam                        0.4424     0.1810
vanilla_lsgan_adam                      0.4547     0.1957
vanilla_lsgan_rmsprop                   0.9206     0.0844
vanilla_lsgan_sgd                       0.9815     0.0302
dcgan_lsgan_sgd                         0.9991     0.0010
dcgan_bce_rmsprop                       1.0089     0.5426
dcgan_bce_adam                          1.1336     0.5168
vanilla_bce_adam                        1.1466     0.6003
vanilla_bce_sgd                         2.8606     0.1896
vanilla_bce_rmsprop                     3.1371     0.2252
dcgan_bce_sgd                          10.1939     0.0001
```

View run **dcgan_wgan_rmsprop** at: https://wandb.ai/mayankjain_25rc006-delhi-technological-university/exp9-gans/runs/vtdel48n
View project at: https://wandb.ai/mayankjain_25rc006-delhi-technological-university/exp9-gans
Synced 5 W&B file(s), 0 media file(s), 0 artifact file(s) and 0 other file(s)
Find logs at: ./wandb/run-20260416_090836-vtdel48n/logs

Upload to HuggingFace

```
HF_REPO = "Mayank2424/exp9-gans"
create_repo(HF_REPO, exist_ok=True)
api = HfApi()

for name in saved_generators.keys():
    for suffix in ['_G.pt', '_D.pt']:
        fname = f'{name}{suffix}'
        if os.path.exists(fname):
            api.upload_file(
                path_or_fileobj=fname,
                path_in_repo=f'models/{fname}',
                repo_id=HF_REPO
```